



# ULX-D

## Wireless Microphone System

Manual for Shure ULXD Single wireless system, including ULXD4, ULXD1, and ULXD2. Includes setup instructions, specifications, and troubleshooting.

Version: 6.8 (2026-I)

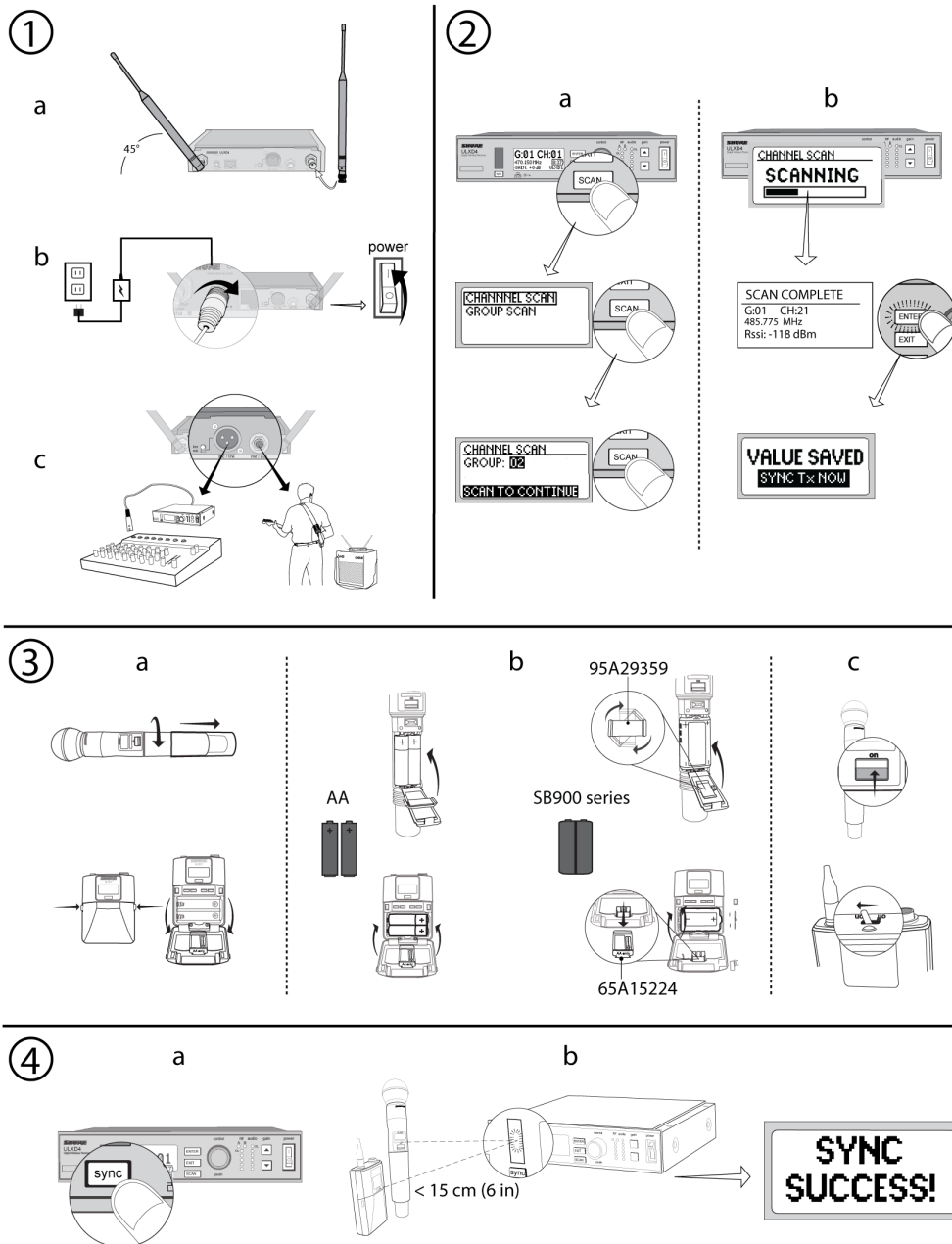
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# ULX-D Wireless Microphone System

## Quickstart Instructions



# General Description

Shure ULX-D™ Digital Wireless offers uncompromising 24-bit audio quality and RF performance, with intelligent, encryption-enabled hardware, flexible receiver options, and advanced rechargeability options for professional sound reinforcement.

A breakthrough in wireless audio quality, Shure digital processing enables ULX-D to deliver the purest reproduction of source material ever available in a wireless system, with a wide selection of trusted Shure microphones to choose from. Extended 20 Hz – 20 kHz frequency range and flat response captures every detail with clarity, presence, and incredibly accurate low end and transient response. With greater than 120 dB, ULX-D delivers wide dynamic range for excellent signal-to-noise performance. Optimized for any input source, ULX-D eliminates the need for transmitter gain adjustments.

ULX-D sets a new and unprecedented standard for spectral efficiency and signal stability. The intermodulation performance of ULX-D is an incredible advancement in wireless performance, enabling a dramatic increase in the number of simultaneous active transmitters on one TV channel. Rock-solid RF signal with zero audio artifacts extends over the entire range. For applications where secure wireless transmission is required, ULX-D offers Advanced Encryption Standard (AES) 256-bit encrypted signal for unbreakable privacy.

Advanced Lithium-ion rechargeability provides extended transmitter battery life over alkaline batteries, battery life metering in hours and minutes accurate to within 15 minutes, and detailed tracking of battery health status.

Generations ahead of any other available system in its class, ULX-D brings a new level of performance to professional sound reinforcement.

## Features

### Uncompromising Professional Digital Wireless

- 24-bit/48 KHz digital audio that exceeds all other systems in accurate reproduction of the source material
- 20 Hz – 20 KHz frequency range with flat response
- Greater than 120 dB dynamic range for excellent signal-to-noise performance
- Advanced Encryption Standard (AES-256) for any application where secure transmission is needed:
  - AES algorithm used in Shure ULX-D™ Wireless has been validated (validation no: 2552) by the National Institute of Standards and Technology (NIST) as conforming to the Advanced Encryption Standard (AES) Algorithm, as specified in Federal Information Processing Standard (FIPS) Publication 197, Advanced Encryption Standard
  - Added benefit of eliminating stray RF interference by allowing only encrypted signal through to the receiver
- Built-in limiter circuitry prevents digital audio clipping from excessive signal levels.
- 60 dB of adjustable system gain easily accessible from the receiver front panel
- No transmitter gain adjustments needed - optimized for any input source
- Wide selection of trusted Shure Microphones

### Extremely Efficient and Reliable RF Performance

- Up to 64 MHz overall tuning range (region dependent)
- Minimized intermodulation distortion results in significantly more channels on air, setting the bar for wireless performance in crowded RF environments
  - 17 active transmitters in one 6 MHz TV channel
  - 22 active transmitters in one 8 MHz TV channel
  - High Density mode enables up to 47 active transmitters in one 6 MHz TV channel (63 in one 8 MHz TV channel), with no audio quality degradation
  - Over 60 compatible channels on one frequency band
- Rock-solid signal stability with no audio artifacts extends over the entire 100 meter line-of-sight range using standard supplied dipole antennas
- Selectable 1, 10, and 20 mW transmitter RF output power (region dependent)

- Optimized scanning automatically finds, prioritizes, and selects the cleanest frequencies available

## Rugged, Intelligent Hardware

- Optimized scanning automatically finds and deploys the cleanest frequencies available
- Proprietary Shure Gain Ranging optimizes system dynamic range for any input source, eliminating the need for transmitter gain adjustments
- Up to 60 dB of adjustable system gain is easily accessible from the receiver front panel
- Networkable receiver simplifies setup across multiple channels
- Compatible with external controllers such as AMX and Creston. Cross-subnet control enabled for management between multiple facilities or venues.
- Rugged metal housing on both transmitters and receiver
- Interference detection and alerts provide instant confirmation when interference is present
- Upgraded LCD screens with adjustable contrast and brightness on the receivers

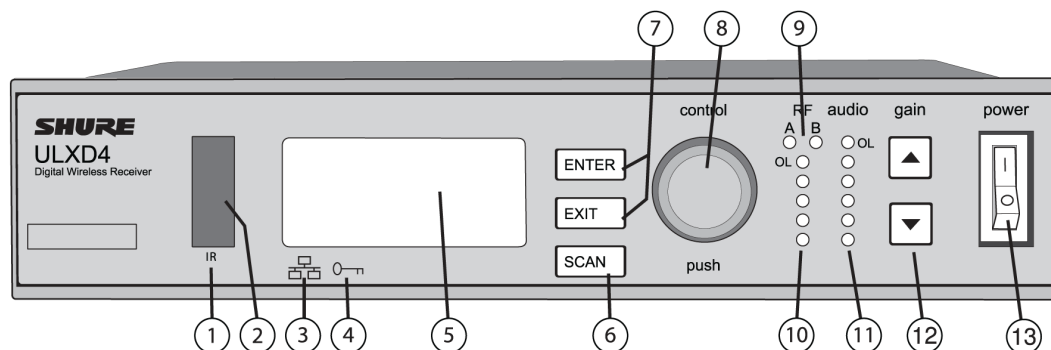
## Advanced Rechargeability - Shure Rechargeable Battery

- Lithium-Ion chemistry and intelligent Shure battery circuitry results in rechargeable batteries with no memory effect.
- The battery can be recharged at any time; a complete discharge is never necessary
- Adapted from industry-leading Axient<sup>®</sup> rechargeable technology
- Lithium-Ion chemistry and intelligent Shure battery circuitry results in rechargeable batteries with zero memory effect and precision metering
- AA alkaline backwards compatibility

# Hardware Interface

## Receiver

### Front Panel



#### ① Sync Button

Press the sync button while the receiver and transmitter IR windows are aligned to transfer settings from the receiver to the transmitter

#### ② Infrared (IR) Sync Window

Sends IR signal to the transmitter for sync

### ③ Network Icon

Illuminates when the receiver is connected with other Shure devices on the network. IP Address must be valid to enable networked control

### ④ Encryption Icon

Illuminates when AES-256 encryption is activated: **Utilities > Encryption**

**Note:** Encryption is not configurable for all units.

### ⑤ LCD Panel

Displays settings and parameters

### ⑥ Scan Button

Press to find the best channel or group

### ⑦ Menu Navigation Buttons

Use to select and navigate through parameter menus

### ⑧ Control Wheel

Push to select menu items for editing, turn to edit a parameter value

### ⑨ RF Diversity LEDs

Indicate antenna status:

- Blue = normal RF signal between the receiver and transmitter
- Red = interference detected
- Off = No RF connection between the receiver and transmitter

**Note:** the receiver will not output audio unless one blue LED is illuminated

### ⑩ RF Signal Strength LEDs

Indicate the RF signal strength from the transmitter:

- Amber = Normal (-90 to -70 dBm)
- Red = Overload (greater than -25 dBm)

### ⑪ Audio LEDs

Indicate average and peak audio levels:

| LED        | Audio Signal Level | Description       |
|------------|--------------------|-------------------|
| Red (6)    | -0.1 dBFS          | Overload/ limiter |
| Yellow (5) | -6 dBFS            | Normal peaks      |
| Yellow (4) | -12 dBFS           |                   |
| Green (3)  | -20 dBFS           | Signal Present    |

| LED       | Audio Signal Level | Description |
|-----------|--------------------|-------------|
| Green (2) | -30 dBFS           |             |
| Green (1) | -40 dBFS           |             |

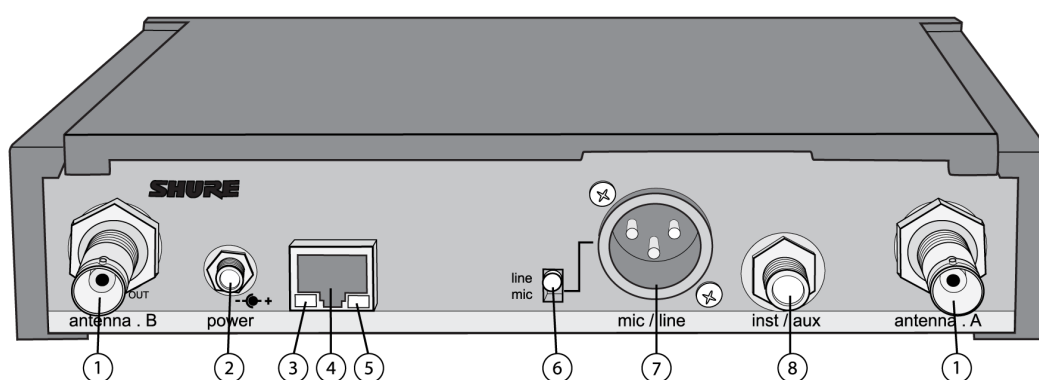
## ⑫ Gain Buttons

Adjust channel gain

## ⑬ Power Switch

Powers the unit on or off

## Back Panel



## ① RF Antenna Diversity Input Jack (2)

For antenna A and antenna B.

## ② Power Supply Jack

Connect the supplied 15 V DC external power supply

## ③ Network Speed LED (Amber)

- Off = 10 Mbps
- On = 100 Mbps

## ④ Ethernet Port

Connect to an Ethernet network to enable remote control and monitoring

**Note:** Use shielded Cat5e or better Ethernet cables for VHF (V50 and V51) to ensure reliable network performance.

## ⑤ Network Status LED (Green)

- Off = no network link
- On = network link active
- Flashing = network link active, flash rate corresponds to traffic volume

## ⑥ Mic/Line Switch

Applies a 30 dB pad in mic position (XLR output only)

## ⑦ **Balanced XLR Audio Output**

Connect to a mic or line level input

## ⑧ **Balanced 1/4" (6.35 mm) TRS Audio Output**

Connect to a mic or line level input

# Transmitters

## ① **Power LED**

- Green = unit is powered on
- Red = low battery or battery error (see Troubleshooting)
- Amber = power switch is disabled

## ② **On/Off Switch**

Powers the unit on or off.

## ③ **SMA Connector**

Connection point for RF antenna.

## ④ **LCD Display:**

View menu screens and settings. Press any control button to activate the backlight.

## ⑤ **Infrared (IR) Port**

Align with the receiver IR port during an IR sync for automated transmitter programming.

## ⑥ **Menu Navigation Buttons**

Use to move through parameter menus and change values.

|              |                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------|
| <b>exit</b>  | Acts as a 'back' button to return to previous menus or parameters without confirming a value change |
| <b>enter</b> | Enters menu screens and confirms parameter changes                                                  |
| <b>▼ ▲</b>   | Use to scroll through menu screens and to change parameter values                                   |

## ⑦ **Battery Compartment**

Requires Shure rechargeable battery or 2 AA batteries.

## ⑧ **AA Battery Adapter**

- Handheld: rotate and store in the battery compartment to use a Shure rechargeable battery
- Bodypack: remove to accommodate a Shure rechargeable battery

## ⑨ **Bodypack Antenna**

For RF signal transmission.

## ⑩ Integrated Antenna

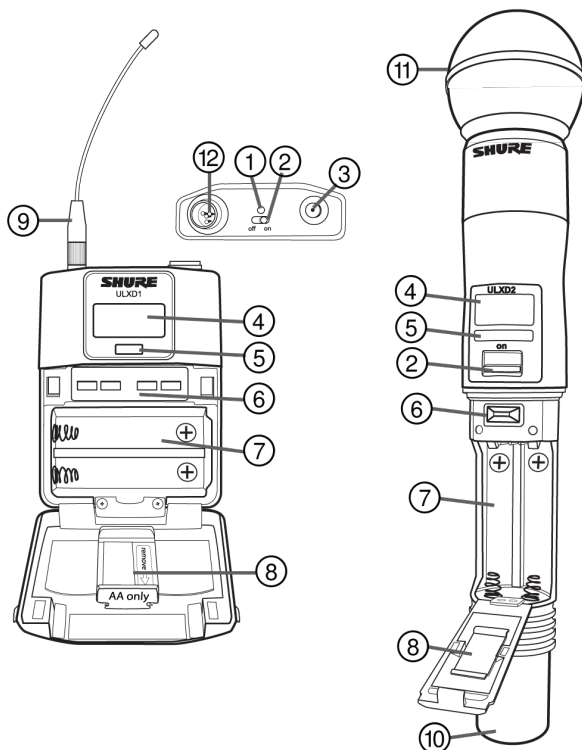
For RF signal transmission.

## ⑪ Microphone Cartridge

See Optional Accessories for a list of compatible cartridges.

## ⑫ TA4M / LEMO Input Jack

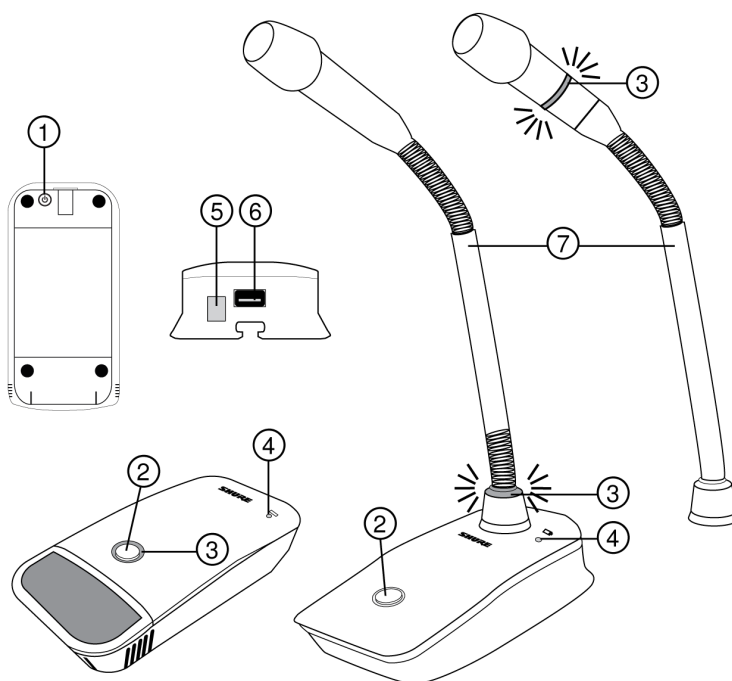
Connects to a microphone or instrument cable.



## Installing Bodypack Antennas

Hand-tighten antennas until secure. Do not use tools.

# Boundary and Gooseneck Base Transmitters



## ① Power Button

Press to power on; press and hold to power off.

## ② Mute/Active Button

Four settings are available for the mute/active button:

- Toggle: Press to switch between active and mute states
- Push-to-Mute: Hold button to mute microphone
- Push-to-Talk: Hold button to activate microphone
- Disabled: Button functionality off

## ③ Mute LED

Indicates whether microphone is active or muted. The following settings are available:

| Active | Muted        |
|--------|--------------|
| Green* | Red*         |
| Red    | Off          |
| Red    | Flashing red |

\*MX400R series gooseneck microphones (red LED) do not offer this setting.

## ④ Low-Battery LED

- Off = More than 30 minutes of battery life remain
- On (red) = Less than 30 minutes of battery life remain

- On (green) = Microphone docked on charging station
- On (amber) = Battery is missing or is not inserted correctly

## ⑤ Infrared (IR) Port

Align with receiver IR port to send settings to transmitter.

## ⑥ Charge Connector

Connects to networked chargers and USB power supply.

## ⑦ Gooseneck Microphone

ULXD8 base fits 5", 10", and 15" Microflex series microphones, available in single or dualflex and with bi-color or red-only LEDs.

# Advanced Transmitter Features

## RF MUTE

Use this to turn on a transmitter without interfering with the RF spectrum.

Press and hold exit during power-on until RF MUTED is displayed. To un-mute, restart the transmitter.



## MIC.OFFSET

MIC.OFFSET compensates for signal level differences between transmitters that share the same receiver channel.

Set the offset gain on a low signal level transmitter to match a louder transmitter: UTILITY > MIC . OFFSET

**Note:** For normal gain adjustments, use the receiver gain buttons.

## Transmitter Audio Mute Mode

Mute Mode reconfigures the transmitter power switch to act a mute switch for the audio. Using the switch, the audio can be easily turned on or muted by presenters, sports referees, or anyone who periodically needs to speak. When the audio is muted, the transmitter RF signal remains on and ready at all times.

**Note:** Mute Mode can be selected as an IR PRESET option.

To set a transmitter to Mute Mode:

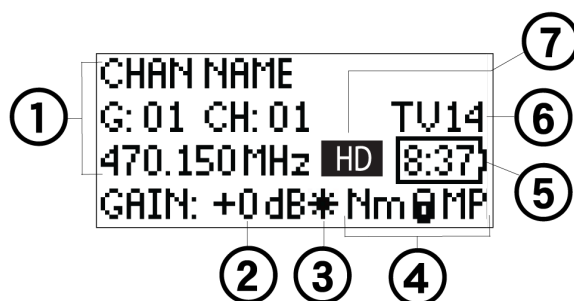
1. From the transmitter menu: UTILITY > MUTE MODE
2. Use the arrows to select ON or OFF.
3. Press enter to save.

**Tip:** The transmitter LED turns red when audio is muted and turns green when audio is enabled. The display of the transmitter will show AUDIO MUTED and the receiver display will show Tx Muted.

**Note:** Mute Mode must be set to OFF in order to use the power switch to turn off the transmitter.

# Menu Screens

## Receiver Channel



### ① Receiver Information

Use DEVICE UTILITIES > HOME INFO to change the home screen display.

### ② Gain Setting

–18 to +42 dB, or Mute.

### ③ Mic. Offset Indicator

Indicates offset gain is added to the transmitter.

### ④ Transmitter Settings

The following information cycles when a transmitter is tuned to the receiver's frequency:

- Transmitter Type
- Input Pad (Bodypack only)
- RF Power Level
- Transmitter Lock Status
- Transmitter Mute Status

### ⑤ Battery Runtime Indicator

Shure rechargeable battery: runtime is displayed in minutes remaining.

AA batteries: runtime is displayed with a 5-bar indicator.

### ⑥ TV Channel

Displays the TV channel that contains the tuned frequency.

### ⑦ High Density Mode Icon

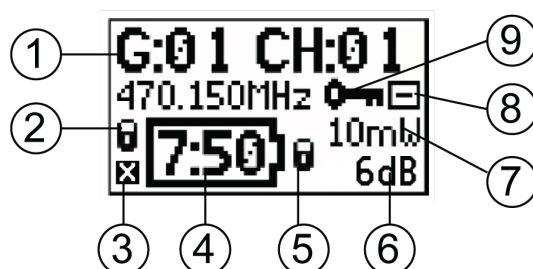
Displayed when High Density mode is enabled.

## Transmitter Status

The following text or icons report transmitter status to the receiver screen:

| Display Icon   | Transmitter Status                                                             |
|----------------|--------------------------------------------------------------------------------|
|                | Bodypack input is attenuated 12 dB                                             |
|                | Offset gain is added to the transmitter                                        |
| <b>Lo</b>      | 1 mW RF power level                                                            |
| <b>Nm</b>      | 10 mW RF power level                                                           |
| <b>Hi</b>      | 20 mW RF power level                                                           |
| <b>M</b>       | Menu is locked                                                                 |
| <b>P</b>       | Power is locked                                                                |
| <b>TxMuted</b> | Displayed when the transmitter audio is set to off using the MUTE MODE feature |
| <b>-No Tx-</b> | No RF connection between a receiver and transmitter or transmitter OFF         |

## Transmitter



### ① Transmitter Information

Scroll ▲ ▼ at the home screen to change the display

### ② Power Lock Indicator

Indicates power switch is disabled

### ③ Transmitter Audio Muted Indicator

Displayed when the transmitter audio is set to off using the MUTE MODE feature.

### ④ Battery Runtime Indicator

- Shure rechargeable battery: runtime is displayed in hours:minutes remaining
- AA Batteries: runtime is displayed with a 5-bar indicator

### ⑤ Menu Lock Indicator

Indicates menu navigation buttons are disabled

**⑥ Mic. Offset**

Displays microphone offset gain value

**⑦ RF Power**

Displays RF power setting

**⑧ Bodypack Input Pad**

The input signal is attenuated 12 dB

**⑨ Encryption Icon**

Indicates encryption is enabled on the receiver and has been transferred to the transmitter from a sync

## Adjusting Receiver Display Brightness and Contrast

Adjust BRIGHTNESS and CONTRAST settings to improve visibility in challenging lighting environments.

1. From the receiver menu: DEVICE UTILITIES > DISPLAY
2. Press the control wheel to select CONTRAST or BRIGHTNESS.
3. Turn the control to adjust the selected parameter.
4. Press ENTER to save changes.

## Home Screen Display Options

### Receiver

The HOME INFO menu provides options to change the information shown on the receiver home screen:

DEVICE UTILITIES > HOME INFO

Use the control wheel to select one of the following screen displays.



CHANNAME  
470.150MHz 8:37  
+10dB ULXD1

Receiver home screen view 1



470.150MHz  
G: 01 CH: 01  
GAIN: +0dB No Tx

Receiver home screen view 2

G:01 CH:01  
 470.150MHz 8:37  
 GAIN: +0dB ULXD1

Receiver home screen view 3

## Transmitter

**Home Screen:** Press the ▲ ▼ arrows at the home menu to display one of the following screens:

CHAN NAME  
 470.150 MHz  
 7:50 10mW

Transmitter home screen view 1

G:01 CH:01  
 470.150MHz  
 7:50 10mW  
 12dB

Transmitter home screen view 2

470.150MHz  
 G:01 CH:01  
 7:50 10mW

Transmitter home screen view 3

## Editing Receiver Channel Name

To edit a receiver channel name, choose EDIT NAME from the menu.

- Turn the control wheel to edit a highlighted character
- Press the control wheel to advance to the next character
- Press ENTER to save changes

**Note:** The channel name is transferred to a transmitter during a sync.

## Setting Regional TV Format

To ensure accurate display of TV channel information, set the TV FORMAT to match the TV channel bandwidth in the region where the receiver is operating. TV bandwidth varies globally, so check local regulations to determine the regional TV bandwidth.

The following TV FORMAT options are available:

- 6 MHz
- 7 MHz
- 8 MHz
- 6 MHz JAPAN
- NO TV (use to turn off TV channel display or in regions where TV channels are not applicable)

To set the TV FORMAT:

1. Menu: DEVICE UTILITIES > ADVANCED RF > TV FORMAT
2. Use the control wheel to select a TV FORMAT option.
3. Press ENTER to save.

## Locking Controls and Settings



Use the LOCK feature to prevent accidental or unauthorized changes to the hardware. Attempting to access a locked feature will display the following message:

### Receiver

Menu path: UTILITIES > LOCK

MENU: All menu paths are inaccessible. To unlock, press the EXIT button, turn the control wheel to select UNLOCKED and press ENTER to save.

GAIN: Gain adjustment is locked

POWER: Power switch is disabled

SCAN: Cannot perform a group scan

SYNC: Cannot sync devices

# Transmitter

Menu path: UTILITY > LOCK

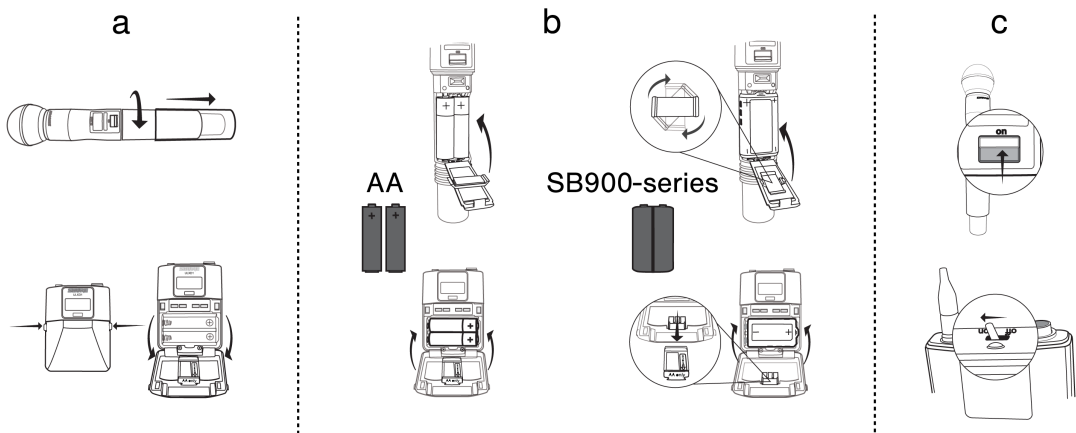
MENU: All menu paths are inaccessible. To unlock, press the ENTER button four (4) times to pass through the following screens: UTILITY > LOCK > MENU UNLOCK

POWER: Power switch is disabled

**Quick-Lock Option:** To turn on the transmitter with its power and menu navigation buttons locked, press and hold the ▲ button during power-on until the locked message is displayed.

To unlock, turn the power switch to the off position, then press and hold the ▲ button while turning the power switch to the on position.

# Batteries



The transmitter runs on two AA batteries or a Shure SB900 series rechargeable battery. Use the included AA battery adapter when using batteries other than the Shure rechargeable battery.

Bodypack: Remove the adapter when using the Shure rechargeable battery

Handheld: Rotate and store the adapter in battery door when using Shure rechargeable battery

# Battery Indicator

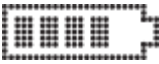
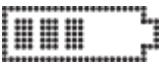
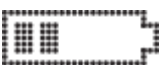
A 5-segment icon on the receiver and transmitter menu screens indicates battery charge.

For accurate battery runtime monitoring, set the transmitter to the appropriate battery type: UTILITY > BATTERY > SET . AA . TYPE.

This table displays the approximate hours and minutes remaining (h:mm).

# Alkaline

| Battery Indicator | UHF/VHF      |              | ISM/1.x      |              |
|-------------------|--------------|--------------|--------------|--------------|
|                   | 1/10 mW      | 20 mW        | 1/10mW       | 20mW         |
|                   | 9:00 to 6:30 | 5:45 to 4:15 | 7:30 to 5:30 | 5:30 to 4:00 |

| Battery Indicator                                                                 | UHF/VHF      |              | ISM/1.x      |              |
|-----------------------------------------------------------------------------------|--------------|--------------|--------------|--------------|
|                                                                                   | 1/10 mW      | 20 mW        | 1/10mW       | 20mW         |
|  | 6:30 to 4:00 | 4:15 to 3:00 | 5:30 to 3:30 | 4:00 to 2:45 |
|  | 4:00 to 1:45 | 3:00 to 1:30 | 3:30 to 1:45 | 2:45 to 1:30 |
|  | 1:45 to 0:45 | 1:30 to 0:30 | 1:45 to 0:45 | 1:30 to 0:30 |
|  | <0:45        | <0:30        | <0:45        | <0:30        |
|  | Replace      | Replace      | Replace      | Replace      |

## Shure SB900 Series Rechargeable Battery

When using a Shure rechargeable battery, the receiver and transmitter home screens display the number of hours and minutes remaining.

Detailed information for the rechargeable battery is displayed in the receiver BATTERY INFO menu and the transmitter menu: UTILITY > BATTERY > BATT. STATS

HEALTH: Displays battery health as a percentage of the charge capacity of a new battery.

CHARGE: Percentage of a full charge

CYCLES: Number of times the battery has been charged

TEMP: Battery temperature in Celsius and Fahrenheit

**Note:** For additional rechargeable battery information, visit [www.shure.com](http://www.shure.com).

```

HEALTH: 90%
CHARGE: 80%
CYCLES: 100
TEMP: 10°C / 50°F

```

## Important Tips for Care and Storage of Shure Rechargeable Batteries

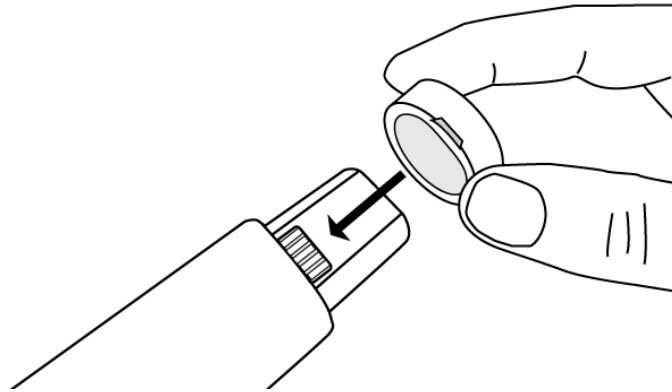
Proper care and storage of Shure batteries results in reliable performance and ensures a long lifetime.

- Always store batteries and transmitters at room temperature
- Ideally, batteries should be charged to approximately 40% of capacity for long-term storage
- During storage, check batteries every 6 months and recharge to 40% of capacity as needed

## Installing the Battery Contact Cover

Install the included battery contact cover (65A15947) on the handheld transmitter to prevent light reflection in broadcast and performance situations.

1. Align the cover as shown.
2. Slide the cover over the battery contacts until it is flush with the transmitter body.



**Note:** Slide the cover off before inserting the transmitter in the battery charger.

## Audio

### Setting Receiver Gain

The receiver gain control sets the audio signal level for the entire receiver and transmitter system. Changes to the gain settings occur in realtime allowing for adjustments during live performances. When adjusting the gain, monitor the audio meter levels to prevent signal overloads.

#### Receiver Gain Controls

The gain can be adjusted by using the gain ▲ ▼ buttons or by entering the AUDIO menu and using the control wheel.

**Tip:** To quickly adjust the gain, press and hold a gain button to enable accelerated scrolling.

#### Reading the Audio Meter



The audio meter displays yellow, green, and red LEDs to indicate the audio signal level. Audio peaks illuminate the LEDs for 2 seconds, while the RMS signal is displayed in realtime.

When setting up the receiver, adjust the gain so that the average signal LED levels are solid green and occasionally yellow, with only the highest peaks causing the red LED to illuminate.

**Tip:** If a vocalist is overloading a bodypack transmitter, try lowering the receiver gain. If additional attenuation is needed, use the transmitter menu to set the INPUT PAD to -12dB.

**Note:** Illumination of the red OL (overload) LED indicates the internal limiter is engaged to prevent digital clipping.

## Muting a Receiver Channel Audio Output

The audio output of each receiver channel can be independently muted to prevent audio from passing. Mute status is indicated by Rx MUTED message appearing on the receiver display in place of the gain value.

**Note:** Receiver gain is disabled for muted channels to prevent unexpected changes in audio levels.

To set a receiver channel output to mute:

1. AUDIO > MUTE
2. Use the control wheel to select ON or OFF.
3. Press ENTER to save.

To unmute the receiver output:

Simultaneously press the ▲ ▼ buttons or select OFF from the MUTE menu option.

**Tip:** Audio mute can be enabled remotely from Wireless Workbench or from an external controller.

**Important!** A power cycle will reset the receiver and unmute the audio output.

## Transmitter Input Clip

The following warning displays on the receiver LCD panel when the transmitter input is clipped:



To correct, select INPUT PAD from the main menu and attenuate the input signal by 12 dB. If the warning persists, reduce the level of input signal.

## Audio Signal Encryption

When encryption is enabled, the receiver generates a unique encryption key which is shared with a the transmitter during an IR sync. Transmitters and receivers that share an encryption key form a protected audio path, preventing unauthorized access from other receivers.

**Note:** For some units, encryption is always on and cannot be configured.

### Encrypting a Single Transmitter to a Single Receiver

1. From the receiver menu: DEVICE UTILITIES > ENCRYPTION > ON (Auto)
2. Press ENTER.
3. Perform an IR Sync to share the encryption key with the selected transmitter.

### Encrypting Multiple Transmitters to a Single Receiver

Multiple transmitters can share the same encryption key, allowing them access to a single receiver. Use this method if you have multiple instruments or wish to use a combination of handheld and bodypack transmitters.

1. From the receiver menu: DEVICE UTILITIES > ENCRYPTION > ON (Manual) > KEEP KEYS.
2. Press ENTER.
3. Perform an IR Sync to share the encryption key with the first transmitter.
4. Turn off the transmitter and perform an IR Sync to share the key additional transmitters.

**Caution!** Make sure only one transmitter is turned on during an IR sync or a performance to avoid causing cross interference between transmitters.

## Regenerating Encryption Keys

Periodically regenerating the encryption key maintains security for transmitters and receivers that are paired for extended periods.

1. From the receiver menu: DEVICE UTILITIES > ENCRYPTION > ON (Manual) > REGENERATE KEYS.
2. Press ENTER.
3. Perform an IR Sync to share the encryption key with the first transmitter.
4. Turn off the transmitter and perform an IR Sync to share the key additional transmitters.

**Caution!** Make sure only one transmitter is turned on during an IR sync or a performance to avoid causing cross interference between transmitters.

## Removing Encryption

**Note:** For some units, encryption is always on and cannot be configured.

1. From the receiver menu: DEVICE UTILITIES ENCRYPTION OFF
2. Press ENTER.
3. IR Sync the transmitter and receiver to clear the encryption key.

**Note:** If multiple transmitters are encrypted to a single receiver, each transmitter must be IR synced to clear the encryption key.

## Scan and Sync

Use this procedure to tune a receiver and transmitter to the best open channel.

**Note:** If you're using transmitters and receivers with mixed bands, such as a wide tuning receiver and a legacy transmitter, see [Mixed Band Operation](#) for instructions.

**Important!** Before you begin:

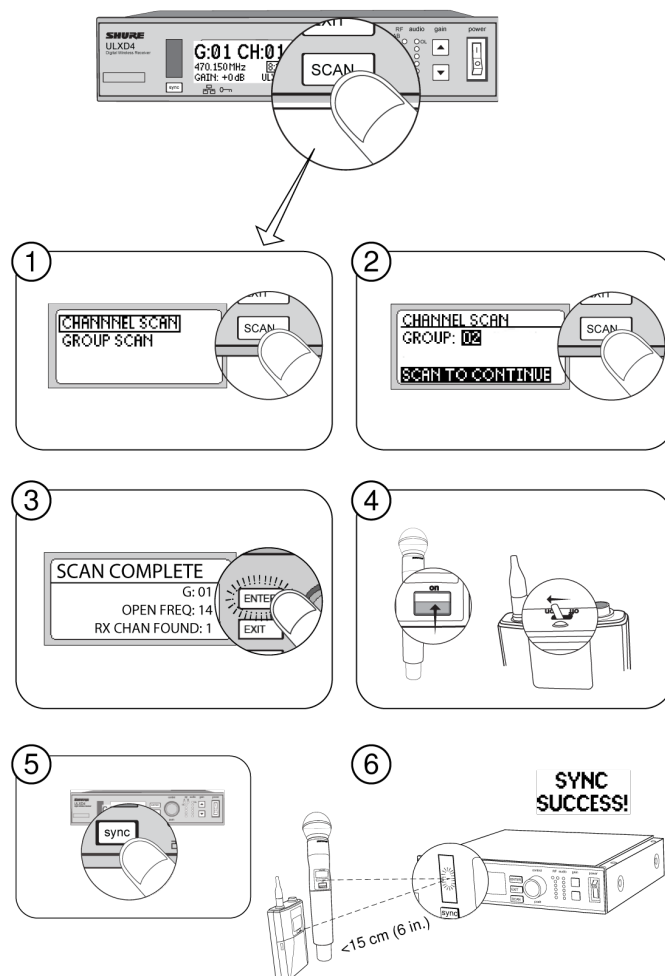
**Turn off** all transmitters to prevent them from interfering with the frequency scan.

**Turn on** any devices that might produce interference during the show so the scan can detect and avoid them, including:

- Other wireless systems or devices
- Computers
- Large LED panels
- Effects processors

1. Perform a group scan on the receiver: SCAN > GROUP SCAN.
2. Press SCAN. While the scan is running, SCANNING appears on the screen.
3. When the scan is done, the receiver displays the group with the most available frequencies. Press the flashing ENTER button to deploy frequencies to each receiver channel.
4. Power on the ULXD transmitter.
5. Press the sync button on the receiver.
6. Align the IR windows until the receiver IR port turns red.

**Note:** When complete, SYNC SUCCESS! appears. The transmitter and receiver are now tuned to the same frequency.



## Multiple System Setup

A setup using networked receivers is the fastest and easiest way to distribute the best open channel to each system. See Networking ULX-D Receivers for networking details.

**Note:** Networked receivers must all be within the same frequency band.

## Networked Receivers

1. Turn on all receivers.
2. Conduct a group scan on the first receiver to find available frequencies in each group: SCAN > GROUP SCAN.
3. Press ENTER to accept the group number and automatically assign the next best channel to each receiver on the network. The receiver LEDs will flash when a frequency has been assigned.
4. Turn on a transmitter and sync to the receiver.

**Important!** Leave the transmitter on and repeat this step for each additional system.

## Non-networked Receivers

1. Turn on all receivers.
2. Conduct a group scan on the first receiver to find available frequencies in each group: SCAN > SCAN > GROUP SCAN > SCAN

- When the scan is complete, use the control wheel to scroll through each group. Press ENTER to select a group that has enough available frequencies for all channels in the system.
- Sync a transmitter to each receiver channel.

**Important!** Leave all transmitters on use the following steps to set up additional receiver channels:

- Set each additional receiver channel to the same group as the first receiver: RADIO > G:
- Conduct a channel scan to find available frequencies within the group: SCAN > SCAN > CHANNEL SCAN > SCAN
- When the scan is complete, press ENTER to assign frequencies to each receiver channel.
- Sync a transmitter to each receiver channel.

## Manual Frequency Selection

To manually adjust group, channel or frequency:

- Select Radio Frequency from the receiver menu.
- Use the control knob to select and adjust the group (G), channel (Ch), or frequency (MHz).
- Press ENTER to save.

## Transmitter IR Presets

Use the IR PRESETS receiver menu to quickly configure transmitter settings from the receiver screen. When a sync is performed between the receiver and transmitter, the IR PRESETS automatically configure the transmitter. Each parameter has the default value KEEP, which leaves that setting unaffected by a sync.

| Feature     | Setting                                        |
|-------------|------------------------------------------------|
| BP PAD      | +0 dB, -12 dB                                  |
| LOCK        | Power, Menu, All, None                         |
| RF POWER    | 10mW=Nm (normal), 1mW=Lo (low), 20mW=Hi (high) |
| BATT        | Alkaline, NiMH, Lithium                        |
| BP OFFSET   | 0 dB to +21 dB (in 3 dB increments)            |
| HH OFFSET   | 0 dB to +21 dB (in 3 dB increments)            |
| MUTE MODE   | OFF, ON                                        |
| Cust. Group | OFF, ON                                        |

**Note:** When Cust. Groups is set to on, it may take up to 30 seconds to complete an IR sync.

## Creating a System Preset

System Presets allow a current receiver setup to be saved and restored. Presets store all receiver settings to provide a quick way to configure a receiver or switch between several different setups. Up to 4 presets can be stored in receiver memory.

To save the current receiver setup as a new preset: DEVICE UTILITIES > SYSTEM RESET > SAVE > CREATE NEW PRESET

Use the control wheel to name the preset, and then press Enter to save.

To recall a saved preset: DEVICE UTILITIES > SYSTEM RESET > RESTORE

Use the control wheel to select the preset name, and then press Enter.

## Mixed-Band Operation

As long as ULX-D transmitters and receivers can be tuned to the same frequency, they can be used together. Mixed-band operation is necessary in order to use wide tuning ULX-D transmitters and receivers with legacy ULX-D transmitters and receivers.

To sync legacy and wide tuning devices, follow these steps to tune them to different, but compatible, bands:

1. Press SCAN on the front of the receiver twice.
2. Choose Channel Scan or Group Scan.
  - If using channel scan: Refer to the tables below to select the group that is compatible with your transmitter band. Press ENTER to complete the scan.
  - If using group scan: After the scan is complete, refer to the tables below and select the group that is compatible with your transmitter band.
3. Press ENTER to deploy frequencies to each receiver channel.
4. Power on the ULXD transmitter.
5. Press the sync button on the receiver.
6. Align the IR windows until the receiver IR port turns red. When complete, SYNC DONE! appears. The transmitter and receiver are now tuned to the same frequency.

**Note:** The transmitter may indicate "G:-- Ch:--" because the 2 RF bands do not share the exact same group-channel map, but the frequency will match and the devices will work together.

### Compatible Groups by Band

#### G54 Receiver - Standard Mode

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
| <b>G52</b>  | 06                            |
|             | 07                            |
|             | 08                            |
| <b>H52</b>  | 09                            |
|             | 10                            |
|             | 11                            |

#### G54 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| <b>G54 HD</b>          | AA                            |
| <b>G52 HD</b>          | AB                            |
| <b>H52 HD</b>          | AC                            |
| <b>TV22</b>            | AD                            |
| <b>TV23</b>            | AE                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| TV24                   | AF                            |
| TV25                   | AG                            |
| TV26                   | AH                            |
| TV27                   | AI                            |
| TV28                   | AJ                            |
| TV29                   | AK                            |
| TV30                   | AL                            |
| TV31                   | AM                            |
| TV32                   | AN                            |

### G55 Receiver - Standard Mode

| Legacy Band       | Group on Wide Tuning Receiver |
|-------------------|-------------------------------|
| G50               | 06                            |
|                   | 07                            |
|                   | 08                            |
| H50               | 09                            |
|                   | 10                            |
|                   | 11                            |
| J50, without TV37 | 12                            |
|                   | 13                            |
|                   | 14                            |

### G55 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| G55 HD, no TV37        | AA                            |
| G50 HD                 | AB                            |
| H50 HD                 | AC                            |
| J50 HD, no TV37        | AD                            |
| TV14                   | AE                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| TV15                   | AF                            |
| TV16                   | AG                            |
| TV17                   | AH                            |
| TV18                   | AI                            |
| TV19                   | AJ                            |
| TV20                   | AK                            |
| TV21                   | AL                            |
| TV22                   | AM                            |
| TV23                   | AN                            |
| TV24                   | AO                            |
| TV25                   | AP                            |
| TV26                   | AQ                            |
| TV27                   | AR                            |
| TV28                   | AS                            |
| TV29                   | AT                            |
| TV30                   | AU                            |
| TV31                   | AV                            |
| TV32                   | AW                            |
| TV33                   | AX                            |
| TV34                   | AY                            |
| TV35                   | AZ                            |
| TV36                   | BA                            |
| TV38                   | BB                            |
| TV39                   | BC                            |
| TV40                   | BD                            |
| TV41                   | BE                            |

## G56 Receiver - Standard Mode

| Legacy/TV Channel Band                                    | Group on Wide Tuning Receiver |
|-----------------------------------------------------------|-------------------------------|
| G51                                                       | 06                            |
|                                                           | 07                            |
|                                                           | 08                            |
| H51                                                       | 09                            |
|                                                           | 10                            |
|                                                           | 11                            |
| UK TV38 with 500 kHz guard band                           | 12                            |
| Belgium 1 TV27+29 w/100 kHz guard band max channels       | 13                            |
| Belgium 2 TV27+29 w/100 kHz guard band standard           | 14                            |
| Belgium 3 TV27+29 w/100 kHz guard band robust             | 15                            |
| EU except UK no TV38 with 100 kHz guard band max channels | 16                            |
| EU except UK no TV38 with 100 kHz guard band standard     | 17                            |

## G56 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| G56 HD                 | AA                            |
| G51 HD                 | AB                            |
| H51 HD                 | AC                            |
| TV21                   | AD                            |
| TV22                   | AE                            |
| TV23                   | AF                            |
| TV24                   | AG                            |
| TV25                   | AH                            |
| TV26                   | AI                            |
| TV27                   | AJ                            |
| TV28                   | AK                            |
| TV29                   | AL                            |
| TV30                   | AM                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| TV31                   | AN                            |
| TV32                   | AO                            |
| TV33                   | AP                            |
| TV34                   | AQ                            |
| TV35                   | AR                            |
| TV36                   | AS                            |
| TV37                   | AT                            |
| TV38                   | AU                            |
| TV39                   | AV                            |
| TV40                   | AW                            |
| TV41                   | AX                            |

### G57 Receiver - Standard Mode

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
| G50         | 06                            |
|             | 07                            |
|             | 08                            |
| H50         | 09                            |
|             | 10                            |
|             | 11                            |
| J50A        | 12                            |
|             | 13                            |
|             | 14                            |

### G57 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| G57 HD                 | AA                            |
| G50 HD                 | AB                            |
| H50 HD                 | AC                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| J50A HD                | AD                            |
| TV14                   | AE                            |
| TV15                   | AF                            |
| TV16                   | AG                            |
| TV17                   | AH                            |
| TV18                   | AI                            |
| TV19                   | AJ                            |
| TV20                   | AK                            |
| TV21                   | AL                            |
| TV22                   | AM                            |
| TV23                   | AN                            |
| TV24                   | AO                            |
| TV25                   | AP                            |
| TV26                   | AQ                            |
| TV27                   | AR                            |
| TV28                   | AS                            |
| TV29                   | AT                            |
| TV30                   | AU                            |
| TV31                   | AV                            |
| TV32                   | AW                            |
| TV33                   | AX                            |
| TV34                   | AY                            |
| TV35                   | AZ                            |
| TV36                   | BA                            |

## G65 Receiver - Standard Mode

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
| G51         | 06                            |

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
|             | 07                            |
|             | 08                            |
|             | 09                            |
| H51         | 10                            |
|             | 11                            |

## G65 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| G56 HD                 | AA                            |
| G51 HD                 | AB                            |
| H51 HD                 | AC                            |
| TV21                   | AD                            |
| TV22                   | AE                            |
| TV23                   | AF                            |
| TV24                   | AG                            |
| TV25                   | AH                            |
| TV26                   | AI                            |
| TV27                   | AJ                            |
| TV28                   | AK                            |
| TV29                   | AL                            |
| TV30                   | AM                            |
| TV31                   | AN                            |
| TV32                   | AO                            |
| TV33                   | AP                            |
| TV34                   | AQ                            |
| TV35                   | AR                            |
| TV36                   | AS                            |
| TV37                   | AT                            |

## G66 Receiver - Standard Mode

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
| G51         | 06                            |
|             | 07                            |
|             | 08                            |
| G52         | 09                            |
|             | 10                            |
|             | 11                            |
| H51         | 12                            |
|             | 13                            |
|             | 14                            |

## G66 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| G66 HD                 | AA                            |
| G51 HD                 | AB                            |
| G52 HD                 | AC                            |
| H51 HD                 | AD                            |
| TV23                   | AE                            |
| TV24                   | AF                            |
| TV25                   | AG                            |
| TV26                   | AH                            |
| TV27                   | AI                            |
| TV28                   | AJ                            |
| TV29                   | AK                            |
| TV30                   | AL                            |
| TV31                   | AM                            |
| TV32                   | AN                            |
| TV33                   | AO                            |
| TV34                   | AP                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| TV35                   | AQ                            |
| TV36                   | AR                            |
| TV37                   | AS                            |

## H54 Receiver - Standard Mode

| Legacy Band | Group on Wide Tuning Receiver |
|-------------|-------------------------------|
| H51         | 06                            |
|             | 07                            |
|             | 08                            |
| L51         | 09                            |
|             | 10                            |
|             | 11                            |

## H54 Receiver - HD Mode

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| H54 HD                 | AA                            |
| H51 HD                 | AB                            |
| L51 HD                 | AC                            |
| TV27                   | AD                            |
| TV28                   | AE                            |
| TV29                   | AF                            |
| TV30                   | AG                            |
| TV31                   | AH                            |
| TV32                   | AI                            |
| TV33                   | AJ                            |
| TV34                   | AK                            |
| TV35                   | AL                            |
| TV36                   | AM                            |
| TV37                   | AN                            |

| Legacy/TV Channel Band | Group on Wide Tuning Receiver |
|------------------------|-------------------------------|
| TV38                   | AO                            |
| TV39                   | AP                            |
| TV40                   | AQ                            |
| TV41                   | AR                            |
| TV42                   | AS                            |
| TV43                   | AT                            |

## RF

### Transmitter RF Power

Reference the following table for setting RF Power:

| RF Power Setting | System Range     | Application                                               |
|------------------|------------------|-----------------------------------------------------------|
| 1 mW             | 33 m (100 ft.)   | For increased channel reuse at close distances            |
| 10 mW            | 100 m (330 ft.)  | Typical setups                                            |
| 20 mW            | >100 m (330 ft.) | For hostile RF environments or long-distance applications |

**Note:** Using the 20 mW setting decreases the transmitter battery runtime and reduces the number of compatible systems.

### Interference Detection



Interference Detection monitors the RF environment for potential sources of interference which can cause audio dropouts.

When interference is identified, the RF LEDs illuminate red and the following warning displays on the receiver LCD panel.

If the warning display persists or the audio drops out repeatedly, perform a Scan and Sync at the first opportunity to find a clear frequency.

## Advanced RF

### RF Mute



Use this to turn on a transmitter without interfering with the RF spectrum.

Press and hold the exit button during power-on until RF MUTED is displayed. To un-mute, restart the transmitter.

### Custom Groups

Use this feature to create up to six groups of manually selected frequencies that can be exported to networked receivers to simplify system setup.

To create a custom group: UTILITIES > ADVANCED RF > CUSTOM GROUPS

**Note:** Use Wireless Workbench or Wireless Frequency Finder to select the best compatible frequencies. See [www.shure.com](http://www.shure.com) for more information.

To export a custom group:

1. Go to UTILITIES > ADVANCED RF > CUSTOM GROUPS > EXPORT. The following screen will display.
2. Press the flashing ENTER button to export all custom groups to all receivers on the network.

### Antenna Bias

Antenna ports A and B provide a DC bias to power active antennas. Set the DC power to off when using passive (non-powered) antennas.

To turn bias off: DEVICE UTILITIES > ADVANCED RF > ANTENNA BIAS > OFF

## Networking Receivers

The receiver uses an Ethernet connection to network with other components. For automatic network configuration, use a DHCP enabled Ethernet switch such as the Shure AXT620. Use multiple Ethernet switches to extend the network for larger installations.

**Note:** Note use only one DHCP server per network.

### Automatic IP Addressing

Set the IP Mode to Automatic for all receivers: DEVICE UTILITIES > NETWORK > CTRL NETWORK

### Manual IP Addressing

1. Connect the receivers to an Ethernet switch.
2. Set the IP Mode to Manual for all devices (DEVICE UTILITIES > NETWORK > CTRL NETWORK)

3. Set valid IP addresses for all devices.
4. Set the subnet mask to the same value for all devices.

## Device ID

Set the receiver Device ID for easy identification within the network or in Wireless Workbench: DEVICE UTILITIES > NET - WORK > SHURE CONTROL > Dev . ID

Use the control wheel to edit the device ID.

## High Density Mode

High Density mode creates additional bandwidth for more channels in crowded RF environments. Frequency efficiency is optimized by running at 1 mW RF transmit power and narrowing the modulation bandwidth, allowing for the channel spacing to be reduced from 350 kHz to 125 kHz. Transmitters can be positioned on adjacent channels with unsubstantial intermodulation distortion (IMD).

High Density mode is ideal for applications where many channels are needed in a confined area, transmission distances are short, and the number of available frequencies is limited. Up to 30 meters of range is available in High Density mode.

## Setting the Receiver to High Density Mode

To set the receiver to High Density mode:

DEVICE UTILITIES > ADVANCED RF > HIGH DENSITY

Use the control wheel to set HIGH DENSITY to ON.

When prompted, sync the transmitter and receiver to enable HIGH DENSITY mode.

**Note:** When the receiver is in HIGH DENSITY mode, the following indicators are shown on the receiver display:

- The HD icon will appear on the receiver display
- The receiver band name will be shown with an "HD" added. (example: The G50 band will appear as G50HD)
- The transmitter group and channel are assigned letters instead of numbers (example: G:AA CH:AA)

## Best Practices for High Density Mode

- When band planning, position ULX-D High Density channels in a range of frequencies separated from other devices.
- Use a separate RF zone for ULX-D High Density channels to prevent intermodulation distortion from other devices.
- During High Density channel scanning, turn on all other transmitters and move them to their intended position.
- Perform a walk test to verify transmitter range
- If using custom groups, the groups loaded into the receiver must be compatible with High Density mode

---

## System Reset

System Reset clears the current receiver settings and restores the factory default settings.

To restore factory default settings:

1. Go to DEVICE UTILITIES > SYSTEM RESET > RESTORE.
  2. Scroll to the DEFAULT SETTINGS option and press ENTER.
  3. Press the flashing ENTER button to return the receiver to the default settings.
-

# Update ULXD Firmware

Often, the first step to troubleshooting an issue with ULXD transmitters or receivers is to perform a firmware update.

Find the current firmware version in the Device Utilities menu on the receiver.

- Receiver firmware version: Device Utilities > Version
- Transmitter firmware version: Device Utilities > Tx FW Update

To check for and install available updates, use [Shure Update Utility](#).

**Note:** If your receiver firmware is older than 2.0.28, refer to [ULX-D Firmware Update and Issues](#) for specific instructions.

For more detailed network setup instructions and information about firmware, refer to the [Shure Update Utility user guide](#).

## Firmware Versions

When you update receiver firmware, update your transmitters to the same firmware version to ensure consistent operation.

The firmware of all ULX-D devices has the form of MAJOR.MINOR.PATCH (e.g., 1.2.14). At a minimum, all ULX-D devices on the network (including transmitters), must have the same MAJOR and MINOR firmware version numbers (e.g., 1.2.x).

## Update the Receiver

First, use Shure Update Utility to send both receiver and transmitter firmware updates to the receiver.

**CAUTION!** Ensure that receiver power and network connections are maintained during a firmware update. Do not turn off the receiver until the update is complete.

**IMPORTANT!** Receivers running firmware prior to 1.4.8 must be updated to 1.4.8 before updating to newer versions. Receivers running firmware between 1.4.8 and 2.0.28 must be updated to 2.0.28 before updating to newer versions. See the [ULX-D Firmware Update and Issues FAQ](#) for more information on this update procedure.

1. Open Shure Update Utility.
2. Click Check Now or [#] updates available to view new versions available for download.
3. Select the updates and click Download.
4. Connect the receiver and computer to the same network.
5. Select a firmware package from the Version to install dropdown menu. Since multiple transmitter types are compatible with ULX-D receivers, firmware packages are named with keys to distinguish which transmitters they support.
  - Keys for firmware that supports ULXD1 and ULXD2 transmitters: ULX-DTx, ULX-D1GTx, ULX-D1smTx
  - Keys for firmware that supports ULXD8 and ULXD8 transmitters: ULX-DBGTx, ULX-DBG1GTx, ULX-DBGTx1sm
6. From the Update Devices tab, send the latest firmware to the receiver.

For more information on the Shure Update Utility, refer to the [Shure Update Utility user guide](#).

## Update the Transmitter

1. Place the transmitter on a microphone or music stand, and roughly align the stand so that the transmitter's IR window can be aligned with the receiver. Shure does not recommend holding the transmitter during the update.
2. To upload the firmware to the transmitter, go to DEVICE UTILITIES > TX FW UPDATE on the receiver.
3. Align the IR ports.
4. Press ENTER on the receiver to begin the download to the transmitter. IR ports must be aligned for the entire download, which can take 50 seconds or longer.

# Shure Wireless Coordination and Management Tools

Use Wireless Workbench and Wireless Workbench mobile to help manage, organize, and coordinate frequencies for your wireless systems.

## Wireless Workbench

Shure Wireless Workbench is a free software application designed to help you manage the RF spectrum and control wireless devices.

- Download Wireless Workbench: [shu.re/wwb](http://shu.re/wwb)
- Get help with Wireless Workbench: [shu.re/wwb-help](http://shu.re/wwb-help)

## Wireless Workbench Mobile

Wireless Workbench Mobile brings wireless monitoring and control to your iOS or Android mobile device in a streamlined format built for field use. Use alongside Wireless Workbench or as a stand-alone application for less complex coordination needs.

- Download Wireless Workbench Mobile: <https://www.shure.com/en-US/docs/guide/wireless-workbench-mobile>
- Get help with Wireless Workbench Mobile: [https://www.shure.com/en-US/products/software/wireless\\_workbench\\_mobile](https://www.shure.com/en-US/products/software/wireless_workbench_mobile)

## Troubleshooting

| Issue                                                                                      | See Solution...      |
|--------------------------------------------------------------------------------------------|----------------------|
| <b>No sound</b>                                                                            | Power, Cables, or RF |
| <b>Faint sound or distortion</b>                                                           | Gain                 |
| <b>Lack of range, unwanted noise bursts, or dropouts</b>                                   | RF                   |
| <b>Cannot turn transmitter off or change frequency settings, or can't program receiver</b> | Interface locks      |
| <b>Encryption error message</b>                                                            | Encryption Mismatch  |
| <b>Transmitter battery hot message</b>                                                     | Tx Battery hot       |

## Power

Make sure that the receiver and transmitter are receiving sufficient voltage. Use the 15 V DC power supply furnished with the ULXD4 receiver. Check the battery indicators and replace the transmitter batteries if necessary.

## Gain

Adjust the system gain on the front of the receiver. Ensure the output level (XLR output only) on the back of the receiver corresponds to the input of the mixing console, amplifier, or DSP.

## Cables

Check that all cables and connectors are working correctly.

## Interface Locks

The transmitter and the receiver can be locked to prevent accidental or unauthorized changes. A locked feature or button will produce the Locked screen on the LCD panel.

## Encryption Mismatch

Re-sync the receiver and transmitter after enabling or disabling encryption.

**Note:** Encryption is not configurable for all units.

## Tx Battery Hot

If the transmitter battery does not cool off, the transmitter will shut down. Let the device cool down and then consider swapping the transmitter battery to continue operation. Identify any possible external heat sources to the transmitter and operate the transmitter away from those external heat sources. All batteries should be stored and operated away from external heat sources in reasonable temperature conditions for best performance.

## Radio Frequency (RF)

### RF LEDs

If neither blue RF Diversity LED is illuminated, then the receiver is not detecting the presence of a transmitter.

The amber RF Signal Strength LEDs indicate the amount of signal being received. This signal could be from the transmitter, **or it could be from an interfering source, such as a television broadcast**. If more than one or two of the amber RF LEDs are still illuminated while the transmitter is off, then that channel has too much interference, and you should try a different channel.

The red RF LED indicates RF overload. This will usually not cause a problem unless you are using more than one system at the same time, in which case, it can cause interference **in the other system**.

## Compatibility

- Perform a Scan and Sync to ensure the transmitter and receiver are set to the same group and channel.
- Look at the label on the transmitter and receiver to make sure they are in the same band (G50, J50, L50, etc...).

## Reducing Interference

- Perform a group or channel scan to find the best open frequency. Perform a sync to transfer the setting to the transmitter.
- For multiple systems, check that all systems are set to channels in the same group (systems in different bands do not need to be set to the same group).
- Maintain a line of sight between transmitter and receiver antennas.
- Move receiver antennas away from metal objects or other sources of RF interference (such as CD players, computers, digital effects, network switches, network cables and Personal Stereo Monitor (PSM) wireless systems).
- Eliminate RF overload (see below).

## Increasing Range

If the transmitter is more than 6 to 60 m (20 to 200 ft) from the receiver antenna, you may be able to increase range by doing one of the following:

- Reduce interference (see above).
- Increase transmitter RF power level.
- Use an active directional antenna, antenna distribution system, or other antenna accessory to increase RF range.

## Eliminating RF Overload

If you see the red RF LED on a receiver, try the following:

- Reduce the transmitter RF power level
- Move the transmitter further away from the receiver—at least 6 m (20 ft)
- If you are using active antennas, reduce antenna or amplifier gain.
- Use omnidirectional antennas

## Contact Customer Support

Didn't find what you need? [Contact our customer support](#) to get help.

## Accessories

### Furnished Accessories

#### All Systems

|                                         |          |
|-----------------------------------------|----------|
| <b>Receiver</b>                         | ULXD4    |
| <b>Power Supply</b>                     | PS43     |
| <b>1/2-Wave Receiver Antennas (2)</b>   | 95T9279  |
| <b>22 in. BNC-BNC Coaxial Cable (2)</b> | 95K2035  |
| <b>Bulkhead Adapter (2)</b>             | 95A8994  |
| <b>Rack-mounting hardware kit</b>       | 90AZ8100 |
| <b>Ethernet Cable, 3' (1)</b>           | 95B15103 |
| <b>AA alkaline batteries (2)</b>        | 80B8201  |

#### Handheld Systems

|                             |                   |
|-----------------------------|-------------------|
| <b>Handheld Transmitter</b> | ULXD2             |
| <b>Cartridge</b>            | see options below |
| <b>Microphone Clip</b>      | 95T9279           |
| <b>Zipper Bag</b>           | 95B2313           |

Choice of one (1) of the following:

|             |        |
|-------------|--------|
| <b>SM58</b> | RPW112 |
| <b>SM86</b> | RPW114 |

|                      |        |
|----------------------|--------|
| <b>SM87A</b>         | RPW116 |
| <b>Beta 58A</b>      | RPW118 |
| <b>Beta 87A</b>      | RPW120 |
| <b>Beta 87C</b>      | RPW122 |
| <b>KSM8 Nickel</b>   | RPW170 |
| <b>KSM8 Black</b>    | RPW174 |
| <b>KSM9</b>          | RPW184 |
| <b>KSM9HS</b>        | RPW186 |
| <b>NXN8/C Black</b>  | RPW200 |
| <b>NXN8/C Nickel</b> | RPW202 |
| <b>NXN8/S Black</b>  | RPW204 |
| <b>NXN8/S Nickel</b> | RPW206 |

## Bodypack System

|                             |         |
|-----------------------------|---------|
| <b>Bodypack Transmitter</b> | ULXD1   |
| <b>Antenna</b>              | 95G9043 |
| <b>Carrying/Storage Bag</b> | 95A2313 |

Choice of one (1) of the following:

|                                                                                                                                                          |                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Cable, Instrument, 2.5 foot (.75 m), 4 Pin Mini Connector (TA4F) to 1/4-inch Connector.</b>                                                           | WA302                             |
| <b>Instrument Clip-on microphone</b>                                                                                                                     | Beta 98H/C                        |
| <b>Lavalier microphone</b>                                                                                                                               | MX150, MX153, WL183, WL184, WL185 |
| <b>Cable, Instrument, 2-foot (0.7m), 4-pin Mini Connector (TA4F) with Right-Angle 1/4-inch Connector, used with Shure Wireless Bodypack Transmitters</b> | WA304                             |
| <b>Premium Guitar Cable TQG Latching Connector</b>                                                                                                       | WA306                             |
| <b>Premium 1/4" Right-Angle</b>                                                                                                                          | WA307                             |
| <b>LEMO to 1/4"</b>                                                                                                                                      | WA308                             |
| <b>LEMO to 1/4" Right-Angle</b>                                                                                                                          | WA309                             |

## Antennas

| <b>Band</b> | <b>1/2-Wave Receiver Antennas</b> | <b>1/4-Wave Transmitter Antennas</b> |
|-------------|-----------------------------------|--------------------------------------|
|-------------|-----------------------------------|--------------------------------------|

|            |          |                  |
|------------|----------|------------------|
| <b>G50</b> | 95AA9279 | 95G9043 (Yellow) |
| <b>G51</b> | 95AA9279 | 95G9043 (Yellow) |
| <b>G52</b> | 95AA9279 | 95G9043 (Yellow) |
| <b>H51</b> | 95AL9279 | 95D9043 (Gray)   |
| <b>H52</b> | 95AL9279 | 95D9043 (Gray)   |
| <b>J50</b> | 95AK9279 | 95E9043 (Black)  |
| <b>K51</b> | 95AJ9279 | 95E9043 (Black)  |
| <b>L50</b> | 95AD9279 | 95E9043 (Black)  |
| <b>L51</b> | 95AD9279 | 95E9043 (Black)  |
| <b>P51</b> | 95AF9279 | 95F9043 (Blue)   |
| <b>R51</b> | 95M9279  | 95F9043 (Blue)   |
| <b>AB</b>  | 95M9279  | N/A              |
| <b>Q51</b> | 95M9279  | N/A              |

## Optional Accessories for ULXD4

| Accessory                                                              | Part      |
|------------------------------------------------------------------------|-----------|
| Shure Lithium-Ion Rechargeable Battery                                 | SB900C    |
| 8-Bay Shure Battery Charger                                            | SBC800    |
| Dual Docking Charger, Power Supply Not Included                        | SBC200    |
| Boundary Transmitter                                                   | ULXD6     |
| Gooseneck-Base Transmitter                                             | ULXD8     |
| 4-bay networked docking station for gooseneck and boundary microphones | SBC450    |
| 8-bay networked docking station for gooseneck and boundary microphones | SBC850    |
| Carrying Case                                                          | WA610     |
| Y-Cable for bodypack transmitters                                      | AXT652    |
| Antenna UHF-R 470-952 MHz                                              | UA845SWB  |
| Passive Antenna/Splitter Combiner Kit (recommended for 2 receivers)    | UA221     |
| In-Line Antenna Amplifier, 470-698MHz                                  | UA830USTV |
| UHF Antenna Power Distribution Amplifier (USA)                         | UA844SWB  |
| UHF Antenna Power Distribution Amplifier (Europe)                      | UA844SE   |
| In-Line Power Supply                                                   | UABIAT    |

| Accessory                                                                                  | Part     |
|--------------------------------------------------------------------------------------------|----------|
| Front Mount Antenna Kit (Includes 2 Cables And 2 Bulkhead)                                 | UA600    |
| 1/2 Wave Antenna Remote Mount Kit                                                          | UA505    |
| Active Directional Antenna 470-900MHZ                                                      | UA874WB  |
| Passive Directional Antenna 470-952 MHz. Includes 10 foot BNC to BNC cable.                | PA805SWB |
| 2 ft. BNC-BNC Coaxial Cable                                                                | UA802    |
| 6 foot (1.8m) BNC to BNC Coaxial Cable for Remote Antenna Mounting for ULX Wireless System | UA806    |
| 25 ft. BNC-BNC Coaxial Cable                                                               | UA825    |
| 50 ft. BNC-BNC Coaxial Cable                                                               | UA850    |
| 100 ft. BNC-BNC Coaxial Cable                                                              | UA8100   |
| Short Rack Bar                                                                             | 53A8611  |
| Rackmount Bracket, Long                                                                    | 53A38729 |
| Link Bars (Bracket)                                                                        | 53B8443  |
| Single Rack Mount Kit                                                                      | RPW503   |
| Dual Rack Mount Kit                                                                        | RPW504   |
| Black Bodypack Pouch                                                                       | WA582B   |

## Specifications

### ULX-D System Specifications

#### RF Carrier Frequency Range

470–932 MHz, varies by region (See Frequency Range and Output Power table)

#### Working Range

100 m ( 330 ft)

Note: Actual range depends on RF signal absorption, reflection and interference. Note: Actual range depends on RF signal absorption, reflection and interference.

#### RF Tuning Step Size

25 kHz, varies by region

#### Image Rejection

>70 dB, typical

## RF Sensitivity

-98 dBm at  $10^{-5}$  BER

## Latency

<2.9 ms

## Audio Dynamic Range

*A-weighted, typical, System Gain @ +10*

|                                                      |         |
|------------------------------------------------------|---------|
| <b>XLR Analog Output</b>                             | >120 dB |
| <b>Dante Digital Output (ULXD4D and ULXD4Q only)</b> | 130 dB  |

## Total Harmonic Distortion

*-12 dBFS input, System Gain @ +10*

<0.1%

## System Audio Polarity

Positive pressure on microphone diaphragm produces positive voltage on pin 2 (with respect to pin 3 of XLR output) and the tip of the 6.35 mm (1/4-inch) output.

## Operating Temperature Range

-18°C (0°F) to 50°C (122°F)

Note: Battery characteristics may limit this range.

## Storage Temperature Range

-29°C (-20°F) to 74°C (165°F)

Note: Battery characteristics may limit this range.

# Receiver Output Level

The following table describes the typical total system gain from the audio input to the receiver outputs:

## Receiver Output Gain

| <b>Output Jack</b>        | <b>System Gain (gain control = 0dB)</b> |
|---------------------------|-----------------------------------------|
| <b>1/4" TRS</b>           | +18 dB                                  |
| <b>XLR (line setting)</b> | +24 dB                                  |
| <b>XLR (mic setting)</b>  | -6 dB*                                  |

\*This setting matches a typical wired SM58 audio signal level.

# ULXD1

## Mic Offset Range

0 to 21 dB (in 3 dB steps)

## Audio Frequency Response

20 – 20 kHz ( $\pm 1$  dB)

## Battery Type

Shure SB900-series Rechargeable Li-Ion or LR6 AA batteries 1.5 V

## Battery Runtime

@ 10 mW

|                     |           |
|---------------------|-----------|
| <b>Shure SB900C</b> | >11 hours |
| <b>alkaline</b>     | 9 hours   |

See Battery Runtime Chart

## Dimensions

86 mm x 66 mm x 23 mm (3.4 in. x 2.6 in. x 0.9 in.) H x W x D

## Weight

142 g (5.0 oz.), without batteries

## Housing

Cast aluminum

## Audio Input

### Connector

4-Pin male mini connector (TA4M), See drawing for details

### Configuration

Unbalanced

### Impedance

1 M $\Omega$ , See drawing for details

### Maximum Input Level

1 kHz at 1% THD

|                |                    |
|----------------|--------------------|
| <b>Pad Off</b> | 8.5 dBV ( 7.5 Vpp) |
| <b>Pad On</b>  | 20.5 dBV ( 30 Vpp) |

### Preamplifier Equivalent Input Noise (EIN)

System Gain Setting  $\geq$  +20-120 dBV, A-weighted, typical

## RF Output

### Connector

SMA

### Antenna Type

1/4 wave

### Impedance

50  $\Omega$

### Occupied Bandwidth

<200 kHz

### Modulation Type

Shure proprietary digital

### Power

1 mW, 10 mW, 20 mW

See Frequency Range and Output Power table, varies by region

## ULXD2

### Mic Offset Range

0 to 21 dB (in 3 dB steps)

### Audio Frequency Response

Note: Dependent on microphone type

### Battery Type

Shure SB900-series Rechargeable Li-Ion or LR6 AA batteries 1.5 V

### Battery Runtime

@ 10 mW

|                     |           |
|---------------------|-----------|
| <b>Shure SB900C</b> | >11 hours |
| <b>alkaline</b>     | 9 hours   |

See Battery Runtime Chart

### Dimensions

|                              |                                              |
|------------------------------|----------------------------------------------|
| <b>VHF V50 and V51</b>       | 278 mm x 51 mm (10.9 in. x 2.0 in.) L x Dia. |
| <b>Other frequency bands</b> | 256 mm x 51 mm (10.1 in. x 2.0 in.) L x Dia. |

### Weight

|                              |                                     |
|------------------------------|-------------------------------------|
| <b>VHF V50 and V51</b>       | 348 g (12.3 oz.), without batteries |
| <b>Other frequency bands</b> | 340 g (12.0 oz.), without batteries |

### Housing

Machined aluminum

## Audio Input

### Configuration

Unbalanced

### Maximum Input Level

1 kHz at 1% THD 145 dB SPL (SM58), typical

Note: Dependent on microphone type

## RF Output

### Antenna Type

Integrated Single Band Helical

### Occupied Bandwidth

<200 kHz

### Modulation Type

Shure proprietary digital

### Power

1 mW, 10 mW, 20 mW

See Frequency Range and Output Power table, varies by region

## ULXD4

### Dimensions

42 mm x 197 mm x 171 mm (1.65 in. x 7.75 in. x 6.75 in.), H x W x D

### Weight

913 g (2.0 lbs), without antennas

### Housing

Steel

### Power Requirements

15 V DC @ 0.6 A, supplied by external power supply (tip positive)

## RF Input

### Spurious Rejection

>80 dB, typical

### Connector Type

BNC

### Impedance

50  $\Omega$

### Bias Voltage

12 - 13 V DC; 170 mA maximum, per antenna

## Audio Output

### Gain Adjustment Range

-18 to +42 dB in 1 dB steps (plus Mute setting)

### Configuration

|                       |                                                                    |
|-----------------------|--------------------------------------------------------------------|
| <b>1/4" (6.35 mm)</b> | Impedance balanced (Tip = audio, Ring = no audio, Sleeve = ground) |
| <b>XLR</b>            | Balanced (1=ground, 2=audio +, 3=audio -)                          |

### Impedance

|                       |                                       |
|-----------------------|---------------------------------------|
| <b>1/4" (6.35 mm)</b> | 100 $\Omega$ (50 $\Omega$ Unbalanced) |
| <b>XLR</b>            | 100 $\Omega$                          |

### Full Scale Output

|                       |                                               |
|-----------------------|-----------------------------------------------|
| <b>1/4" (6.35 mm)</b> | +12 dBV                                       |
| <b>XLR</b>            | LINE setting = +18 dBV, MIC setting = -12 dBV |

Mic/Line Switch  
30 dB pad

Phantom Power Protection

|                |     |
|----------------|-----|
| 1/4" (6.35 mm) | Yes |
| XLR            | Yes |

Networking

Power Over Ethernet (PoE)  
No, protected

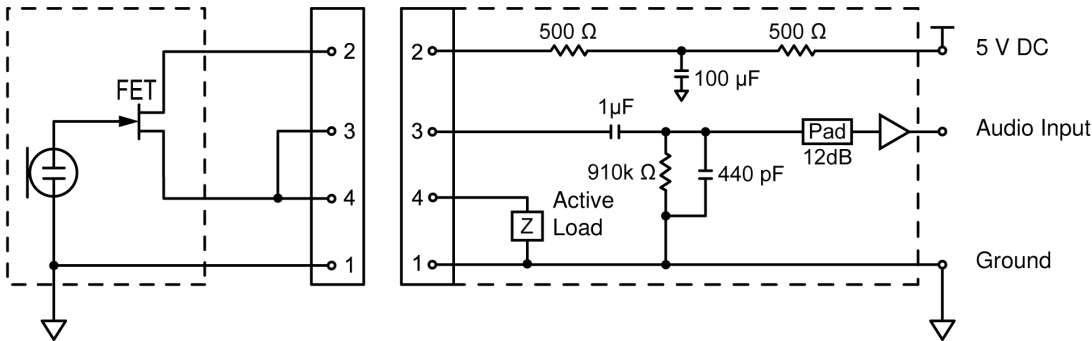
Network Interface  
Single Port Ethernet 10/100 Mbps

Network Addressing Capability  
DHCP or Manual IP address

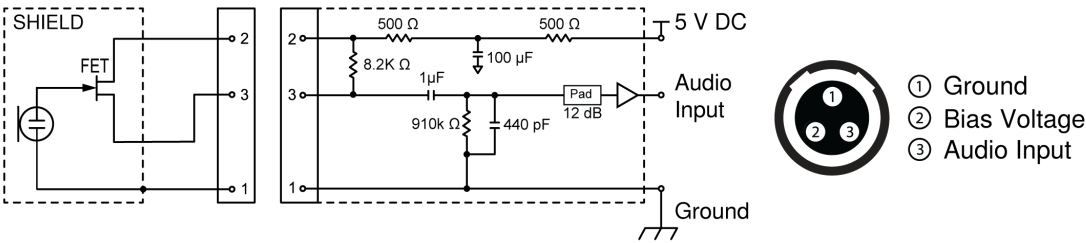
Maximum Cable Length  
100 m (328 ft)

Tables and Diagrams

TA4M Connector

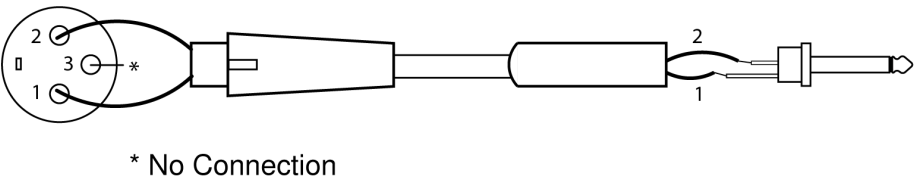


## LEMO Connector



## XLR to ¼ Output

Use the following wiring diagram to convert the XLR output to a ¼ output.



## Batteries

### ULXD Battery Life

| MHz          | hours   |       |          |       |
|--------------|---------|-------|----------|-------|
|              | SB900C  |       | alkaline |       |
|              | 1/10 mW | 20 mW | 1/10 mW  | 20 mW |
| 470 to 810   | >11:30  | >7:40 | >8       | >5:30 |
| 902 to 928   | >10:10  | >6:20 | >7       | >4    |
| 174 to 216   | >11:30  | >8:55 | 8        | >5    |
| 1240 to 1800 | >10:45  | >8:15 | >6       | >4:30 |

**Note:** SB900-series batteries use SBC200, SBC800 and SBC220 chargers.

The values in this table are typical of fresh, high quality batteries. Battery runtime varies depending on the manufacturer and age of the battery.

## Frequency Range and Transmitter Output Power

| Band | Frequency Range ( MHz ) | Power delivered to antenna port ( mW RMS )<br>(Lo/Nm/Hi) |
|------|-------------------------|----------------------------------------------------------|
| G50  | 470 to 534              | 1/10/20                                                  |
| G51  | 470 to 534              | 1/10/20                                                  |

| Band           | Frequency Range ( MHz ) | Power delivered to antenna port ( mW RMS )<br>(Lo/Nm/Hi) |
|----------------|-------------------------|----------------------------------------------------------|
| G52            | 479 to 534              | 1/10                                                     |
| G53            | 470 to 510              | 1/10/20                                                  |
| G54*           | 479 to 565              | 1/10/20                                                  |
| G55*†          | 470 to 608, 614 to 636  | 1/10/20                                                  |
| G56*           | 470 to 636              | 1/10/20                                                  |
| G57*           | 470 to 608              | 1/10/20                                                  |
| G62            | 510 to 530              | 1/10/20                                                  |
| G65*           | 470 to 606              | 1/10/20                                                  |
| G66*           | 487 to 606              | 1/10                                                     |
| H50            | 534 to 598              | 1/10/20                                                  |
| H51            | 534 to 598              | 1/10/20                                                  |
| H52            | 534 to 565              | 1/10                                                     |
| H54*           | 520 to 636              | 1/10/20                                                  |
| J50            | 572 to 636              | 1/10/20                                                  |
| J50A Δ         | 572 to 608              | 1/10/20                                                  |
| J51            | 572 to 636              | 1/10/20                                                  |
| K51            | 606 to 670              | 1/10                                                     |
| L50            | 632 to 696              | 1/10/20                                                  |
| L51            | 632 to 696              | 1/10/20                                                  |
| L53            | 632 to 714              | 1/10/20                                                  |
| M19            | 694 to 703              | 1/10/20                                                  |
| P51            | 710 to 782              | 1/10/20                                                  |
| R51            | 800 to 810              | 1/10/20                                                  |
| JB (Tx only)   | 806 to 810              | 1/10                                                     |
| AB (Rx and Tx) | 770 to 810              | A Band (770-805): 1/10/20                                |
|                |                         | B Band (806-809): 1/10                                   |
| Q12            | 748 to 758              | 1/10/20                                                  |
| Q51            | 794 to 806              | 10                                                       |

| Band | Frequency Range ( MHz )    | Power delivered to antenna port ( mW RMS )<br>(Lo/Nm/Hi) |
|------|----------------------------|----------------------------------------------------------|
| V50  | 174 to 216                 | 1/10/20                                                  |
| V51  | 174 to 216                 | 1/10/20                                                  |
| V52  | 174 to 210                 | 10                                                       |
| X50  | 925 to 932                 | 1/10                                                     |
| X51  | 925 to 937.5               | 10                                                       |
| X52  | 902 to 928                 | 0.25/10/20                                               |
| X53  | 902 to 907.500, 915 to 928 | 0.25/10/20                                               |
| X54  | 915 to 928                 | 0.25/10/20                                               |
| Z16  | 1240 to 1260               | 1/10/20                                                  |
| Z17  | 1492 to 1525               | 1/10/20                                                  |
| Z18  | 1785 to 1805               | 1/10/20                                                  |
| Z19  | 1785 to 1800               | 1/10/20                                                  |
| Z20  | 1790 to 1805               | 1/10/20                                                  |

\*Supported by ULXD1, ULXD2, ULXD4D, and ULXD4Q only

†Operation mode varies according to region. The maximum power level for Peru is 10mW.

△ Output power limited to 10 mW above 608 MHz.

**Note:** Frequency bands might not be available for sale or authorized for use in all countries or regions.

The band Z17 (1492-1525 MHz) must be used indoors only.

For the Band Z19 (1785-1800MHz) used in Australia, per Radio Communications Low Interference Potential Devices Class License 2015; item 30 note C: the system must be operated within the range of 1790-1800MHz when used outdoors.

## Frequencies for European Countries

### G51 470-534 MHz



| Country Code                          | Frequency Range     |
|---------------------------------------|---------------------|
| Code de Pays                          | Gamme de frequences |
| Codice di paese                       | Gamme di frequenza  |
| Código de país                        | Gama de frecuencias |
| Länder-Kürzel                         | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, EST, F   | *                   |
| FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| M, N, NL, P, PL, RO, S, SK, SLO, TR   | *                   |
| All other countries                   | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## G56 470-636 MHz



| Country Code                          | Frequency Range     |
|---------------------------------------|---------------------|
| Code de Pays                          | Gamme de frequences |
| Codice di paese                       | Gamme di frequenza  |
| Código de país                        | Gama de frecuencias |
| Länder-Kürzel                         | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, EST, F   | *                   |
| FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| M, N, NL, P, PL, RO, S, SK, SLO, TR   | *                   |
| all other countries                   | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## H51 534-598 MHz



| Country Code                 | Frequency Range     |
|------------------------------|---------------------|
| Code de Pays                 | Gamme de frequences |
| Codice di paese              | Gamme di frequenza  |
| Código de país               | Gama de frecuencias |
| Länder-Kürzel                | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, EST | 534 - 598 MHz*      |
| F, GB, GR, H, I, IS, L, LT   | 534 - 598 MHz*      |
| NL, P, PL, S, SK, SLO        | 534 - 598 MHz*      |
| DK, FIN, M, N                | *                   |
| HR, E, IRL, LV, RO, TR       | *                   |
| All other countries          | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## K51 606-670 MHz



| Country Code                      | Frequency Range                      |
|-----------------------------------|--------------------------------------|
| Code de Pays                      | Gamme de frequences                  |
| Codice di paese                   | Gamme di frequenza                   |
| Código de país                    | Gama de frecuencias                  |
| Länder-Kürzel                     | Frequenzbereich                      |
| A, B, BG, CH, CY, CZ, D, EST      | 606 - 670 MHz*                       |
| F, GB, GR, H, I, IS, L, LT        | 606 - 670 MHz*                       |
| NL, P, PL, S, SK, SLO             | 606 - 670 MHz*                       |
| RO                                | 646 - 647; 654 - 655; 662 - 663 MHz* |
| DK, E, FIN, HR, IRL, LV, M, N, TR | *                                    |
| All other countries               | *                                    |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## L52 632-694 MHz



| Country Code                          | Frequency Range     |
|---------------------------------------|---------------------|
| Code de Pays                          | Gamme de frequences |
| Codice di paese                       | Gamme di frequenza  |
| Código de país                        | Gama de frecuencias |
| Länder-Kürzel                         | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, EST, F   | *                   |
| FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| M, N, NL, P, PL, RO, S, SK, SLO, TR   | *                   |
| All other countries                   | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## P51 710-782 MHz



| Country Code                               | Frequency Range                                            |
|--------------------------------------------|------------------------------------------------------------|
| Code de Pays                               | Gamme de frequences                                        |
| Codice di paese                            | Gamme di frequenza                                         |
| Código de país                             | Gama de frecuencias                                        |
| Länder-Kürzel                              | Frequenzbereich                                            |
| A, B, BG, CH, CY, CZ, D, EST, F, GB        | 710 - 782 MHz*                                             |
| GR, H, I, IS, L, LT, NL, P, PL, S, SK, SLO | 710 - 782 MHz*                                             |
| RO                                         | 718 - 719; 726 - 727; 734 - 743; 750 - 751; 758 - 759 MHz* |
| DK, E, FIN, HR, IRL, LV, M, N, TR          | *                                                          |
| All other countries                        | *                                                          |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## Q51 794-806 MHz



| Country Code                             | Frequency Range     |
|------------------------------------------|---------------------|
| Code de Pays                             | Gamme de frequences |
| Codice di paese                          | Gamme di frequenza  |
| Código de país                           | Gama de frecuencias |
| Länder-Kürzel                            | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, E, EST      | *                   |
| F, FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| LV, M, N, NL, P, PL, S, SK, SLO, TR      | *                   |
| All other countries                      | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## R51 800-810 MHz



| Country Code                             | Frequency Range     |
|------------------------------------------|---------------------|
| Code de Pays                             | Gamme de frequences |
| Codice di paese                          | Gamme di frequenza  |
| Código de país                           | Gama de frecuencias |
| Länder-Kürzel                            | Frequenzbereich     |
| N                                        | 800 - 810 MHz*      |
| A, B, BG, CH, CY, CZ, D, DK, E, EST      | *                   |
| F, FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| LV, M, N, NL, P, PL, S, SK, SLO, TR      | *                   |
| All other countries                      | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## S50 823-832 MHz, 863-865 MHz



| Country Code                          | Frequency Range     |
|---------------------------------------|---------------------|
| Code de Pays                          | Gamme de frequences |
| Codice di paese                       | Gamme di frequenza  |
| Código de país                        | Gama de frecuencias |
| Länder-Kürzel                         | Frequenzbereich     |
| D                                     | license free        |
| A, B, BG, CH, CY, CZ, D, DK, EST, F   | *                   |
| FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| M, N, NL, P, PL, RO, S, SK, SLO, TR   | *                   |
| 863 - 865 MHz                         | EU: license free    |
| All other countries                   | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## V51 174-216 MHz



| Country Code                             | Frequency Range     |
|------------------------------------------|---------------------|
| Code de Pays                             | Gamme de frequences |
| Codice di paese                          | Gamme di frequenza  |
| Código de país                           | Gama de frecuencias |
| Länder-Kürzel                            | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, E, EST      | *                   |
| F, FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| LV, M, N, NL, P, PL, S, SK, SLO, TR      | *                   |
| All other countries                      | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## Z17 1492-1525 MHz



| Country Code                                                                                  | Frequency Range     |
|-----------------------------------------------------------------------------------------------|---------------------|
| Code de Pays                                                                                  | Gamme de frequences |
| Codice di paese                                                                               | Gamme di frequenza  |
| Código de país                                                                                | Gama de frecuencias |
| Länder-Kürzel                                                                                 | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, E, EST                                                           | *                   |
| F, FIN, GB, GR, H, HR, I, IRL, IS, L, LT                                                      | *                   |
| LV, M, N, NL, P, PL, S, SK, SLO, TR                                                           | *                   |
| This Wireless microphone operates on the range of 1492-1525 MHz. Should be used INDOORS ONLY. |                     |
| All other countries                                                                           | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## Z18 1785-1805 MHz



| Country Code                             | Frequency Range     |
|------------------------------------------|---------------------|
| Code de Pays                             | Gamme de frequences |
| Codice di paese                          | Gamme di frequenza  |
| Código de país                           | Gama de frecuencias |
| Länder-Kürzel                            | Frequenzbereich     |
| A, B, BG, CH, CY, CZ, D, DK, E, EST      | *                   |
| F, FIN, GB, GR, H, HR, I, IRL, IS, L, LT | *                   |
| LV, M, N, NL, P, PL, S, SK, SLO, TR      | *                   |
| All other countries                      | *                   |

\* This equipment may be capable of operating on some frequencies not authorized in your region. See [Licensing Information](#).

## Safety and Regulatory Information for Wireless Products

### Explanation of Symbols

|  |                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------|
|  | This symbol indicates that dangerous voltage constituting a risk of electric shock is present within this unit. |
|--|-----------------------------------------------------------------------------------------------------------------|



This symbol indicates that there are important operating and maintenance instructions in the literature accompanying this unit.

## Important Safety Instructions

1. READ these instructions.
2. KEEP these instructions.
3. HEED all warnings.
4. FOLLOW all instructions.
5. DO NOT use this apparatus near water.
6. CLEAN ONLY with dry cloth.
7. DO NOT block any ventilation openings. Allow sufficient distances for adequate ventilation and install in accordance with the manufacturer's instructions.
8. DO NOT install near any heat sources such as open flames, radiators, heat registers, stoves, or other apparatus (including amplifiers) that produce heat. Do not place any open flame sources on the product.
9. DO NOT defeat the safety purpose of the polarized or grounding type plug. A polarized plug has two blades with one wider than the other. A grounding type plug has two blades and a third grounding prong. The wider blade or the third prong are provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
10. PROTECT the power cord from being walked on or pinched, particularly at plugs, convenience receptacles, and the point where they exit from the apparatus.
11. ONLY USE attachments/accessories specified by the manufacturer.
12. USE only with a cart, stand, tripod, bracket, or table specified by the manufacturer, or sold with the apparatus. When a cart is used, use caution when moving the cart/apparatus combination to avoid injury from tip-over.



13. UNPLUG this apparatus during lightning storms or when unused for long periods of time.
14. REFER all servicing to qualified service personnel. Servicing is required when the apparatus has been damaged in any way, such as power supply cord or plug is damaged, liquid has been spilled or objects have fallen into the apparatus, the apparatus has been exposed to rain or moisture, does not operate normally, or has been dropped.
15. DO NOT expose the apparatus to dripping and splashing. DO NOT put objects filled with liquids, such as vases, on the apparatus.
16. The MAINS plug or an appliance coupler shall remain readily operable.
17. The airborne noise of the Apparatus does not exceed 70dB (A).
18. Apparatus with CLASS I construction shall be connected to a MAINS socket outlet with a protective earthing connection.
19. To reduce the risk of fire or electric shock, do not expose this apparatus to rain or moisture.
20. Do not attempt to modify this product. Doing so could result in personal injury and/or product failure.
21. Operate this product within its specified operating temperature range.
22. Follow local regulations and consult qualified personnel if the product installation or relocation requires construction work. Choose mounting hardware and an installation location that can support the weight of the product. Avoid locations subject to constant vibration. Use the required tools to install the product properly. Inspect the product periodically.

### WARNING:

- Voltages in this equipment are hazardous to life. No user-serviceable parts inside. Refer all servicing to qualified service personnel. The safety certifications do not apply when the operating voltage is changed from the factory setting.
- If water or other foreign objects enter the inside of the device, fire or electric shock may result.

## Safety Information for Batteries

1. Battery packs may explode or release toxic materials. Risk of fire or burns. Do not open, crush, modify, disassemble, heat above 140°F (60°C), or incinerate.
2. Follow instructions from manufacturer
3. Only use Shure charger to recharge Shure rechargeable batteries

**WARNING:**



Danger of explosion if battery incorrectly replaced. Replace only with same or equivalent type.

Do not dispose of the battery along with household waste. Check with local vendor for proper disposal of used battery packs.

4. Never put batteries in mouth. If swallowed, contact your physician or local poison control center
5. In the event of a cell leaking, do not allow the liquid to come in contact with the skin or eyes. If contact has been made, wash the affected area with copious amounts of water and seek medical advice.
6. Do not short circuit; may cause burns or catch fire
7. Do not charge or use battery packs other than Shure rechargeable batteries
8. Batteries (battery pack or batteries installed) shall not be exposed to excessive heat such as sunshine, fire or the like
9. Do not immerse the battery in liquid such as water, beverages, or other fluids.
10. Do not attach or insert battery with polarity reversed.
11. Keep away from small children.
12. Do not use abnormal batteries.
13. Pack the battery securely for transport.

**Note:** Use only with the included power supply, batteries, or a Shure-approved equivalent.

## Storing Batteries

If batteries are to be stored for more than eight days, they should be placed into a temperature controlled storage area. Recommended battery storage temperature is 10 to 25°C (50 to 77°F).

Additional battery storage information is available at [shure.com/battery-storage](https://shure.com/battery-storage).



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## Regulatory Information for Class B EMC Products

### CE Notice

Hereby, Shure Incorporated declares that this product with CE Marking has been determined to be in compliance with European Union requirements.

The full text of the EU declaration of conformity is available at the following site: <https://www.shure.com/en-EU/support/declarations-of-conformity>.

## UKCA Notice

Hereby, Shure Incorporated declares that this product with UKCA Marking has been determined to be in compliance with UK-CA requirements.

The full text of the UK declaration of conformity is available at the following site: <https://www.shure.com/en-GB/support/declarations-of-conformity>.

## Cybersecurity STATEMENT OF COMPLIANCE

**Product Type:** Relevant connectable products defined as internet-connectable products or network-connectable products, in line with inter alia Product Security and Telecommunications Infrastructure Act 2022.

**Manufacturer Statement:** We, Shure Incorporated, certify and declare as manufacturer under our sole responsibility, that the above-mentioned product(s) conform(s) to the legislation as mentioned under Attachment 1 – to Cybersecurity Statement of Compliance listed here: <https://www.shure.com/en-GB/about-us/security>.

**Information on how to report security issues:** The latest version of Shure's Disclosure policy can be found at the following link: <https://www.shure.com/en-GB/about-us/security>

**Security update periods:** Shure provides support regarding hardware and software updates that continue the integral cyber security safety of Shure products up to 24 months after end of life (AEOL). For the full statement regarding Shure's product support policy, and information regarding products end of life status information can be found at the following link: <https://www.shure.com/en-GB/about-us/security>

**Manufacturer:**

Shure Incorporated 5800 Touhy Avenue  
Niles, Illinois, 60714-4608 U.S.A. Website: [www.Shure.com](http://www.Shure.com).

**Technical documentation is kept at:**

Shure Incorporated, Corporate Global Compliance Engineering Division

**UK Importer/Representative:**

Shure UK Limited  
Unit 2, The IO Centre, Lea Road, Waltham Abbey, Essex, EN9 1AS, U.K.  
Phone: +44 (0)1992 - 703058  
Email: [EMEAsupport@shure.de](mailto:EMEAsupport@shure.de)

**On behalf of Manufacturer:**

Chad Ayers  
08 May 2025 Niles, Illinois  
Senior Director, Global Compliance

## FCC Notice

This equipment has been tested and found to comply with the limits for a Class B digital device pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference with radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference with radio or television reception, which can be determined by turning the equipment off and on, you are encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the antenna of the radio/television receiver.
- Increase the separation between this equipment and the radio/television receiver.
- Plug the equipment into a different outlet so that the equipment and the radio/television receiver are on different power mains branch circuits.
- Consult a representative of Shure or an experienced radio/television technician for additional suggestions.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesired operation.

**Notice:** The FCC regulations provide that changes or modifications not expressly approved by Shure Incorporated could void your authority to operate this equipment.

For information regarding responsible party and other matters relating to FCC compliance, please contact Shure Incorporated, 5800 W. Touhy Avenue, Niles, Illinois 60714-4608 U.S.A. [shure.com/contact](https://www.shure.com/contact)

This Class B digital apparatus complies with Canadian ICES-003. Cet appareil numérique de la classe B est conforme à la norme NMB-003 du Canada.

## Canada, ISED Notice

**Notice:** The Industry Canada regulations provide that changes or modifications not expressly approved by Shure Inc. could void your authority to operate this equipment.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. End user must follow the specific operating instructions for satisfying RF exposure compliance. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance 20 cm between the radiator & your body.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. End user must follow the specific operating instructions for satisfying RF exposure compliance. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

The portable device is designed to meet the requirements for exposure to radio waves established by the Federal Communications Commission (USA). These requirements set a SAR limit of 1.6 W/kg averaged over one gram of tissue. The highest SAR value reported under this standard during product certification for use when properly worn on the body/head. This equipment should be installed and operated with minimum distance 0 cm between the radiator and your body/head.

## Canada Warning for Wireless

This device operates on a no-protection, no-interference basis. Should the user seek to obtain protection from other radio services operating in the same TV bands, a radio licence is required. For further details, consult Innovation, Science and Economic Development Canada's document Client Procedures Circular CPC-2-1-28, Voluntary Licensing of Licence-Exempt Low-Power Radio Apparatus in the TV Bands.

Ce dispositif fonctionne selon un régime de non-brouillage et de non-protection. Si l'utilisateur devait chercher à obtenir une certaine protection contre d'autres services radio fonctionnant dans les mêmes bandes de télévision, une licence radio serait

requis. Pour en savoir plus, veuillez consulter la Circulaire des procédures concernant les clients CPC.2.1.28, Délivrance de licences sur une base volontaire pour les appareils radio de faible puissance exempts de licence et exploités dans les bandes de télévision d'Innovation, Sciences et Développement économique Canada.

## Industry Canada (IC) Notices

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

1. L'appareil ne doit pas produire de brouillage;
2. L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

This equipment complies with ISSED radiation exposure limits set forth for an uncontrolled environment. End user must follow the specific operating instructions for satisfying RF exposure compliance. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Cet équipement est conforme aux limites d'exposition aux rayonnements ISSED établies pour un environnement non contrôlé. L'utilisateur final doit suivre les instructions spécifiques pour satisfaire les normes. Cet émetteur ne doit pas être co-implanté ou fonctionner en conjonction avec toute autre antenne ou transmetteur.

This equipment complies with ISSED radiation exposure limits set forth for an uncontrolled environment. End user must follow the specific operating instructions for satisfying RF exposure compliance. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter. The antenna(s) must be installed such that a minimum separation distance of 20 cm is maintained between the radiator (antenna) and all persons at all times.

Cet équipement est conforme aux limites d'exposition aux rayonnements ISSED établies pour un environnement non contrôlé. L'utilisateur final doit suivre les instructions spécifiques pour satisfaire les normes. Cet émetteur ne doit pas être co-implanté ou fonctionner en conjonction avec toute autre antenne ou transmetteur. La ou les antennes doivent être installées de telle façon qu'une distance de séparation minimum de 20 cm soit maintenue entre le radiateur (antenne) et toute personne à tout moment.

This equipment complies with ISSED radiation exposure limits set forth for an uncontrolled environment. This equipment could be installed and operated with minimum distance 0 cm between the radiator and your body. End user must follow the specific operating instructions for satisfying RF exposure compliance. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

The portable device is designed to meet the requirements for exposure to radio waves established by the ISSED. These requirements set a SAR limit of 1.6 W/kg averaged over one gram of tissue. The highest SAR value reported under this standard during product certification for use when properly worn on the body/head.

Cet équipement est conforme aux limites d'exposition aux rayonnements ISSED établies pour un environnement non contrôlé. Cet équipement peut-être installé et utilisé avec une distance minimale de 0 cm entre le radiateur et votre corps. L'utilisateur final doit suivre les instructions spécifiques pour satisfaire les normes. Cet émetteur ne doit pas être co-implanté ou fonctionner en conjonction avec toute autre antenne ou transmetteur.

Le dispositif portatif est conçu pour répondre aux exigences d'exposition aux ondes radio établie par le développement énergétique DURABLE. Ces exigences un SAR limite de 1,6 W/kg en moyenne pour un gramme de tissu. La valeur SAR la plus élevée signalée en vertu de cette norme lors de la certification de produit à utiliser lorsqu'il est correctement porté sur les corps/tête.

Additional Canadian information on RF exposure also can be found at the following Web address: <http://www.ic.gc.ca/eic/site/smt-gst.nsf/eng/sf08792.html>

## ACMA Notice

WARNING: This device operates under an ACMA class license and must comply with all conditions of that license including operating frequencies.

## ANATEL Notice

Este equipamento não tem direito à proteção contra interferência prejudicial e não pode causar interferência em sistemas devidamente autorizados. Para maiores informações, consulte o site da ANATEL – <http://www.anatel.gov.br>.

## IFETEL Notice

La operación de este equipo está sujeta a las siguientes dos condiciones: (1) es posible que este equipo o dispositivo no cause interferencia perjudicial y (2) este equipo o dispositivo debe aceptar cualquier interferencia, incluyendo la que pueda causar su operación no deseada.

## KCC Notice

해당 무선설비는 전파혼신 가능성이 있으므로 인명안전과 관련된 서비스는 할 수 없음

## NCC Notice

Connection and use of this communications equipment is permitted by the Nigerian Communications Commission.

## NCC Notice

低功率射頻器材技術規範

取得審驗證明之低功率射頻器材，非經核准，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。低功率射頻器材之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。前述合法通信，指依電信管理法規定作業之無線電通信。低功率射頻器材須忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。

614MHz-703MHz: 使用頻段供其他通訊業務使用時，器材應即停止使用

## SRRC Notice

- (一) 本产品符合“微功率短距离无线电发射设备目录和技术要求”的具体条款和使用场景；
- (二) 不得擅自改变使用场景或使用条件、扩大发射频率范围、加大发射功率（包括额外加装射频功率放大器），不得擅自更改发射天线；
- (三) 不得对其他合法的无线电台（站）产生有害干扰，也不得提出免受有害干扰保护；
- (四) 应当承受辐射射频能量的工业、科学及医疗（ISM）应用设备的干扰或其他合法的无线电台（站）干扰；
- (五) 如对其他合法的无线电台（站）产生有害干扰时，应立即停止使用，并采取措施消除干扰后方可继续使用；
- (六) 在航空器内和依据法律法规、国家有关规定、标准划设的射电天文台、气象雷达站、卫星地球站（含测控、测距、接收、导航站）等军民用无线电台（站）、机场等的电磁环境保护区域内使用微功率设备，应当遵守电磁环境保护及相关行业主管部门的规定。

# Environmental Regulatory Information

## Waste Electrical and Electronic Equipment (WEEE) Directive



In the European Union and the United Kingdom, this label indicates that this product should not be disposed of with household waste. It should be deposited at an appropriate facility to enable recovery and recycling.

## Registration, Evaluation, Authorization of Chemicals (REACH) Directive

REACH (Registration, Evaluation, Authorization of Chemicals) is the European Union (EU) and the United Kingdom (UK) chemical substances regulatory framework. Information on substances of very high concern contained in Shure products in a concentration above 0.1% weight over weight (w/w) is available upon request.

## Recycling Information

Please consider the environment, electric products and packaging are part of regional recycling schemes and do not belong to regular household waste.

## 中国 RoHS

| 部件名称   | 有害物质 |    |    |        |     |      |     |     |      |      |
|--------|------|----|----|--------|-----|------|-----|-----|------|------|
|        | Pb   | Cd | Hg | Cr(VI) | PBB | PBDE | DBP | BBP | DIBP | DEHP |
| 电路模块   | X    | ○  | ○  | ○      | ○   | ○    | ○   | ○   | ○    | ○    |
| 金属模块   | X    | ○  | ○  | ○      | ○   | ○    | ○   | ○   | ○    | ○    |
| 线缆及其组件 | X    | ○  | ○  | ○      | ○   | ○    | ○   | ○   | ○    | ○    |
| 电源适配器* | X    | ○  | ○  | ○      | ○   | ○    | ○   | ○   | ○    | ○    |
| 锂电池组*  | X    | ○  | ○  | ○      | ○   | ○    | ○   | ○   | ○    | ○    |

注 1: ○: 表示该有害物质在该部件所有均质材料中的含量均不超出电器电子产品有害物质限制使用国家标准要求。  
X: 表示该有害物质至少在该部件某一均质材料中的含量超出电器电子产品有害物质限制使用国家标准要求。  
注 2: 本产品大部分的部件采用无害的环保材料制造, 含有有害物质的部件皆因全球技术发展水平的限制而无法实现有害物质的替代。  
注 3: 以上未列出的部分, 表明其有害物质含量均不超出电器电子产品有害物质限制使用国家标准要求  
\*: 表示如果包含部分

## Certifications

### ULXD1, ULXD2, ULXD1LEMO3

**IC:** 616A-ULXD1 G50, 616A-ULXD1 H50, 616A-ULXD1 J50, 616A-ULXD1 L50; 616A-ULXD2 G50, 616A-ULXD2 H50, 616A-ULXD2 J50, 616A-ULXD2 L50, 616A-ULXD1G50S, 616A-ULXD1H50S, 616A-ULXD1J50AS, 616A-ULXD2G50S, 616A-ULXD2H50S, 616A-ULXD2J50AS, 616A-ULXD1G57, 616A-ULXD2G57.

**FCC:** DD4ULXD1G50, DD4ULXD1H50, DD4ULXD1J50, DD4ULXD1L50; DD4ULXD2G50, DD4ULXD2H50, DD4ULXD2J50, DD4ULXD2L50, DD4ULXD1-G50, DD4ULXD1-H50, DD4ULXD1-J50A, DD4ULXD1-X52, DD4ULXD2-G50, DD4ULXD2-H50, DD4ULXD2-J50A, DD4ULXD2-X52, DD4ULXD1G57, DD4ULXD2G57 .

**IC:** 616A-ULXD1X52, 616A-ULXD2X52

**FCC:** DD4ULXD1X52, DD4ULXD2X52

**IC:** 616A-ULXD1V50, 616A-ULXD2V50

**FCC:** DD4ULXD1V50, DD4ULXD2V50

**Note:** For transmitters operating in the V50 and V51 bands: nominal free space antenna gain at middle of the band is typically -6 dBi, and rolls off at the band edges an additional -4 dB.